

Title

Impact of cholera outbreak response immunization: Insights from modeling outbreaks in sub-Saharan Africa, 2010-2022.

Abstract

Background

Cholera remains a critical public health threat in sub-Saharan Africa, where more than 1,600 outbreaks were recorded between 2010 and 2022. Although oral cholera vaccines (OCVs) is now a cornerstone of outbreak response, delays in their deployment can diminish their impact, raising questions about optimal immunization strategies under resource constraints.

Methods

We modeled the impact of outbreak response immunization (ORI) campaigns for 1,621 cholera outbreaks in sub-Saharan Africa (2010–2022). The analysis accounted for age-specific vaccine effectiveness (direct and indirect), deployment delays, and time to develop immunity. Parameter values were sampled across broad ranges to capture uncertainty. We assessed ORI impact on both individual outbreaks and on a subset of outbreaks (reflecting the number of OCVs deployed since the global stockpile's establishment in 2013). Key outcome measures included (percentage) cases and deaths averted, disability-adjusted life years averted (per 1,000 OCVs), and cost-effectiveness.

Findings

At week 1, the median (percentage) reduction in cases was 133 [95% prediction interval: 0 - 7,187] (70.9% [95% PI: 0% - 95.3%]), but the impact halved for every two weeks of deployments delays or for every 44 additional cases. Vaccine efficiency was about 0.6 cases averted per 1,000 OCVs (costing US\$5,500) at week 1, but it declined rapidly (and costs rose) with deployment delays. Outcomes varied widely across outbreaks, with larger outbreaks deriving greater benefits. When vaccination was initiated at nine weeks—broadly comparable to observed ORI delays—the median reduction for individual outbreaks was 2.8% [95% PI: 0 - 74.3], yet aggregated across multiple outbreaks, it reached 33.4% [95% PI: 19.2 - 45.3]. Strategically targeting larger outbreaks led to greater impact, improved efficiency, and better cost-effectiveness than random allocation.

Conclusion

Timely vaccination is essential to maximize ORI impact, yet even delayed campaigns can still achieve substantial benefits when their impacts are aggregated across multiple outbreaks. Prioritizing larger outbreaks can bolster the overall impact. These findings underscore the potential value of ORI and highlight the need for data-driven frameworks to guide cholera vaccine allocation and control efforts.

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Key messages

1. We assessed the impact of ORI campaigns for cholera using detailed data from 1,621 outbreaks in sub-Saharan Africa between 2010 and 2022.
2. ORI impact in reducing cholera cases declined sharply with deployment delays, with the median case reduction halving for every two-week delay during the first 10 weeks.
3. For a nine-week delay—the median deployment time reported in the Global Task Force on Cholera Control (GTFCC) dashboard—median case reduction dropped to under 5% for individual outbreaks, with substantial variability. However, when the impact was aggregated across sampled outbreaks, ORIs still reduced cases by 33.4% [95% PI: 19.2 - 45.3], owing to the larger impact in high-burden outbreaks.
4. Targeted ORIs focusing on larger and in particular, larger outbreaks could improve the number of cases averted (per 1,000 OCVs) several fold.

Research in context

Evidence before this study

We searched PubMed on March 15, 2024, with no language or date restrictions, using the search terms (("Cholera"[Mesh] OR "Vibrio cholerae"[Mesh])) AND "Cholera Vaccines"[Mesh] AND ((model*[Title/Abstract]) OR (math*[Title/Abstract])) AND (vaccine*[Title/Abstract]), yielding 164 articles. Of these, 17 studies met the inclusion criteria, focusing on cholera transmission modeling to evaluate vaccine impact under various scenarios. Most studies simulated outbreaks in single, unique geographic setting¹⁻¹³, while only a few articles examined outbreaks across more than one geographic setting.¹⁴⁻¹⁷ These studies varied significantly in campaign type, including reactive mass campaigns, proactive campaigns, “hotspot” or area-targeted campaigns. Variability in outbreak size, duration, and modeling parameters—such as vaccine delivery and immunogenicity delays—were also common. For example, while some authors assumed protection immediately following vaccination^{9,13-15}, others considered a lag between vaccination and protection.^{1,6,8,10} Most prior models focused on large outbreaks (e.g., Zimbabwe with 98,591 cases or Haiti with 119,902 cases)^{8,15} and lacked small or short-duration outbreaks. These gaps highlight the need for models that reflect a broader spectrum of outbreak characteristics to enhance the generalizability of vaccine impact assessments.

Added value of this study

To our knowledge, this is the first study to model cholera vaccine impact across a large, diverse dataset of outbreaks from multiple temporal and geographic contexts in sub-Saharan Africa. By analyzing 1,406 unique outbreaks from 2010 to 2023—ranging in size, duration, attack rate, and population dynamics—our model offers a comprehensive and generalizable framework for evaluating oral cholera vaccine (OCV) impact in outbreak response immunizations (ORIs). Unlike previous studies focused on isolated large outbreaks, our approach captures the variability of outbreaks in their responses to vaccination timing and coverage. This broader modeling framework provides insights into cases and deaths averted and the cost-effectiveness of ORI campaigns across different outbreak characteristics and resource allocation strategies.

Implications of the available evidence

Our study offers critical insights for guiding global cholera vaccine implementation strategy. We demonstrate that ORI effectiveness declines by half approximately every two weeks during the first 10 weeks of delay, underscoring the importance of timely campaign implementation. Despite these delays, ORIs remain highly effective for large, prolonged, and high-attack-rate outbreaks, even when initiated as late as nine weeks post-outbreak onset. For example, with a 9-week delay, the median number of cases averted was 6 (<5%) across all outbreaks, whereas the largest 10% of outbreaks saw a median of 1,338 cases averted (>40%). When the vaccine impact is assessed across outbreaks collectively, overall percentage reduction in cases reaches 33.4% [95% prediction interval (PI): 19.2 - 45.3] and 0.63 [95% PI: 0.346 - 0.932] cases are averted per 1,000 OCVs. These impacts can be further amplified if ORI campaigns are prioritized for large outbreaks. These findings highlight the need for targeted vaccination strategies that prioritize high-burden outbreaks (e.g., high attack-rate outbreaks), enabling more efficient resource allocation and maximizing the overall impact of limited OCV supplies.

Main text

Introduction

Cholera, one of the oldest infectious diseases affecting humans, has persisted as a major public health challenge since the 7th pandemic began near the Bay of Bengal in 1961.¹⁸ It remains a significant concern in regions with inadequate water, sanitation, and hygiene (WaSH), particularly in low- and middle-income countries (LMICs).¹⁹ Modelled estimates of cholera incidence and cases reported to the World Health Organization (WHO) indicate that sub-Saharan Africa bears the majority of the global burden, characterized by substantial spatial heterogeneity.^{20–22} For instance, analysis of cholera incidence time series from 2010 to 2016 in sub-Saharan Africa identified approximately 1,000 outbreaks of varying scale across different administrative units.²³ These data also reveal considerable temporal

heterogeneity within regions, alongside spatial variations, likely complicating intervention efforts.

Sustainable access to adequate WaSH, alongside case management and surveillance, remains the foundation of cholera control. However, the oral cholera vaccine (OCV) serves as a critical supplementary tool, especially in areas where other interventions are challenging to implement. The WHO recommends OCV use for controlling endemic cholera and during outbreaks, establishing a global OCV stockpile in 2013 with Gavi's financial support.²⁴ Between 2013 and 2024, over 200 million doses have been deployed from this stockpile for both proactive and reactive vaccination campaigns.²⁵ Rising demand, outpacing global production capacity, prompted the International Coordinating Group (ICG)—which manages the stockpile for WHO and Gavi—to adopt a single-dose regimen for outbreak response in October 2022, replacing the standard two-dose schedule, to address the urgent needs of reactive campaigns.²⁶ This shift underscores the need to optimize OCV impact amid constrained supply.

Given limited resources, policymakers—from local governments to global funders—prioritize efficient OCV use. Studies highlighting spatial heterogeneity and cholera “hotspots” suggest that targeting high-risk areas, identified through covariates, historical incidence²⁶, or sensitive diagnostics,¹⁷ could maximize vaccine benefits. While this approach seems logical, accurately identifying high-risk populations is often hampered by temporal fluctuations in cholera incidence,²⁷ waning vaccine-derived immunity,²⁸ and population mobility.^{9,29} These factors can diminish the effectiveness of targeting, even in areas with a high average risk, particularly when vaccine availability covers only a small fraction of at-risk populations. In contrast, outbreak-response immunizations (ORIs)—vaccinating populations where outbreaks are detected—reduce uncertainties tied to preemptive targeting. In practice, the OCV stockpile has primarily supported ORIs, with over 200 million doses deployed by 2024, of which a substantial proportion (77%) addressed outbreaks rather than prevention.²⁵ Understanding the impact of ORIs, which dominate

vaccination efforts, and identifying strategies to enhance their effectiveness are thus critical.

Existing studies offer valuable insights into the potential impact of preventive campaigns and ORIs, yet they lack comprehensive scope.^{51,3,5,6,10,15,23} Many focus on a handful of large or prolonged outbreaks,^{1,3,5,6,10,15,23} which fail to represent the diverse epidemiological profiles—size, duration, location, and attack rate—of cholera outbreaks, as evidenced by data from sub-Saharan Africa between 2010 and 2023.^{23,30} Additionally, these studies often overlook key operational and immunological factors. For example, while some explore early-stage vaccination, real-world delays exceeding 66 days in emergencies are common,³¹ occasionally rendering vaccines irrelevant if outbreaks subside before deployment. Similarly, assumptions about vaccine protection vary: some studies presume immediate immunity,^{9,13–15} while others account for delays in immunogenicity, reflecting the time required for immunity to develop.^{1,6,8,10} A modeling approach integrating realistic epidemiological diversity, immunological delays, and operational constraints of ORIs is essential to assess their impact and inform global cholera control strategies. This study fills these gaps by analyzing a dataset of over 1,600 cholera outbreaks in sub-Saharan Africa from 2010 to 2023,²³ capturing wide variation in size, duration, attack rate, and spatiotemporal distribution. *Our model incorporates practical ORI details—such as implementation delays and immunological lags in vaccine-induced immunity—and* evaluates diverse timing and coverage scenarios. We also assess the economic implications of ORIs through cost-effectiveness analyses, using a resampled dataset reflecting outbreaks likely to trigger ORIs given limited OCV supply. This comprehensive framework aims to guide strategic vaccine deployment and maximize ORI impact in resource-limited settings.

Methods

Overview

We modeled the potential impact of ORIs on 1,406 cholera outbreaks in sub-Saharan Africa from 2010 to 2023. Our goal was to include as many outbreak data as possible to ensure a comprehensive representation of cholera epidemiology. However, we excluded outbreaks that may have already been influenced by OCVs, as these would not serve as appropriate baseline case series (**Figure 1A**). Our model simulated case and death counts under counterfactual vaccination scenarios that accounted for direct and indirect protective effects of OCV. In addition, we incorporated age-specific vaccine effectiveness, potential delays in vaccine deployment and administration, and the time required for the vaccine-induced immunity to develop (**Figure 1B**). We then compared scenarios to estimate percentage reductions in cases, deaths, and disability-adjusted life years (DALYs). These outcomes were converted to vaccination campaign efficiency metrics (e.g., cases, deaths, and DALYs averted per 1,000 OCVs) and cost-effectiveness metrics (e.g., cost per case, death, or DALY averted). The computer codes and datasets used to replicate the results discussed in the paper can be found in the GitHub repository.³²

Outbreak data

We compiled weekly suspected cholera incidence data from the Cholera Taxonomy Data,³³ an expanded dataset derived from the existing database. This resource encompasses 1,664 outbreaks reported in various administrative units across 30 countries in sub-Saharan Africa from January 2010 to July 2023.^{23,30} Outbreaks were defined by location-specific average weekly incidence thresholds, spanning from the first to the last reported suspected case, with any weeks lacking data assumed to have zero cases.³⁰ Outbreaks that had OCV interventions occurred during or within the preceding three years were excluded, as their trajectories would be influenced by vaccine-derived protection and thus not serve as valid baseline (no vaccination) scenarios.²⁸ Consequently, 1,406 suspected cholera outbreaks were included in our modeling.

Modeling the vaccine impact

We focus on the immediate impact of the ORI during ongoing outbreaks and do not account for subsequent outbreak prevention in the same population. This assumption reflects the expectation that vaccination campaigns are unlikely sustain initial vaccine coverage, given the potentially rapid waning of single-dose regimens, population mobility, and other factors. We estimated the impact of ORIs by calculating the proportion of cases averted through both direct and indirect vaccine effects, in other words, overall effectiveness of the vaccine.³⁴ (See Supplementary Information **S1**). Our model incorporated parameters for vaccination campaign (e.g., age-specific vaccine effectiveness and coverage levels), vaccine immunology (e.g., time required for immunity to develop), cost-effectiveness (**Table 1**).

The proportion of cases averted, $P_{i,\pi}$, at a given vaccine coverage π was calculated for two age groups—children under five ($i = 1$) and individuals aged five and older ($i = 2$)—using:

$$P_{i,\pi} = 1 - [\pi(1 - \psi_i)(1 - \eta_\pi) + (1 - \pi)(1 - \eta_\pi)],$$

where ψ_i represents direct vaccine effectiveness for age group i and η_π denotes the indirect vaccine effectiveness at coverage π . Thus, unvaccinated individuals, comprising $1 - \pi$ of the population, still benefit from indirect protection $1 - \eta_\pi$. We estimated η_π through Bayesian beta regression of incidence rates among unvaccinated individuals in a randomized trial conducted in Matlab, Bangladesh, supplemented by simulations based on the trial data.^{11,35} (Supplementary Information **S2** and **Figure S1**).

The cumulative number of cases, $C_{i,\pi,\tau}$, for age group i , under a vaccination campaign initiated at time τ with coverage π was calculated as:

$$C_{i,\pi,\tau} = \sum_{t=1}^T I_{t,i} [1 - \mathbf{1}(t \geq \tau + \delta) P_{i,\pi}],$$

where $I_{t,i}$ is the incidence at time t for age group i and $\mathbf{1}(x)$ is an indicator function that equals 1 if x is true and 0 otherwise. The term δ captures delays for campaign implementation and the development of vaccine-derived immunity, and T denotes the outbreak's end date.

Overall, we simulated ORIs in each of the 1,406 outbreaks, exploring various vaccination coverage levels, campaign timing, and triggers (time-triggered or case-triggered). We sampled 200 parameter sets via a Sobol low-discrepancy sequence (**Table 2** and Supplementary Information **S3**), yielding varying estimates of ORI impact. Results are reported as medians with 95% prediction intervals (PIs), defined by the central 95% of the simulated distributions. Outbreaks were assessed both individually and in aggregate, using either simple random sampling or weighted sampling based on cumulative cases, attack rate, and outbreak duration.

Our static modeling approach does not explicitly represent dynamic changes in infection states or the force of infection. Instead, we assume that once individuals are vaccinated, vaccine recipients acquire partial immunity, and the cumulative number of cases declines in proportion to $P_{i,\pi}$. This enables an assessment of vaccine's overall impact without the complexity of simulating transmission dynamics for more than 1,400 outbreaks. Our approach aligns with standard vaccine effectiveness estimators in clinical trials, typically computed as 1 - ratio of incidence rate in the vaccinated population to that in the unvaccinated control group.^{36–39} Further, conventional SEIR-type dynamic models (**Figure S2**) often fail to capture the diversity and complexity of cholera outbreaks in the dataset (see **Figure S3** for examples). Nonetheless, we validated our results against those generated by a dynamic modeling approach for a subset of outbreaks where such methods closely replicated the observed outbreak patterns (Supplementary Information **S4**).

Table 1. Model parameters.

Description	Values	Source
<i>Parameters varied</i>		
Direct vaccine effectiveness (i.e., reduction in susceptibility) by one dose of OCV for the < 5 yo and 5+ yo, respectively.	30% [95% CI: 15%, 42%] (< 5 yo) and 64% [95% CI: 58%, 70%] (5+ yo)	28
Campaign duration	5 [range: 3-14] days	31
Indirect vaccine effectiveness	79% [38% – 99%] and 99% [49%-100%] at 60% and 90% vaccine coverages, respectively. (Supplementary Information S3)	
Delay in vaccine-derived protection	10.5 [range: 3, 21] days (Supplementary Information S6)	40–48
<i>Parameters fixed</i>		
Vaccine coverage	60% - 90%	
Proportion of mild, moderate, and severe infection	0.648, 0.289, 0.069	49
Disability weight for mild, moderate, and severe cholera	0.074, 0.188, 0.247	49
Vaccine cost of procurement	USD 1.65	50
Vaccine delivery cost	USD 2.32	51
Vaccine shipping cost	USD 0.06	52
Cost to patient for cholera hospitalization and outpatient services	US\$109.32 and US\$ 16.06	53
Cost to public system for cholera hospitalization and outpatient services	US\$ 184.97 and US\$ 7.99	53

Number of workdays lost to patients and caregivers	5.8 and 3.9 days	53
Mean age of infection	26 years	54–56
Duration of illness	2 days	57
Life expectancy at a given age	Varies by year and country	58
Age distribution of sub-Saharan African countries	Varies by year and country	58
Gross Domestic Product (GDP)	Varies by year and country	59
Population in workforce	Varies by year and country	60

Campaign triggers and assumptions

Vaccination campaigns were launched according to two criteria: (1) the time elapsed since the outbreak's onset (time-triggered) and (2) the cumulative number of reported symptomatic cases (case-triggered) (**Table 2**). The time-triggered approach highlights how campaign delays influence vaccine impact during an outbreak, offering insights into the interplay between timing and vaccine impact. By contrast, the case-triggered approach reflects real-world conditions in which the exact outbreak start date may be uncertain, but the cumulative number of cases by the vaccination date is known. Under the case-triggered approach, the same cumulative-case threshold could trigger earlier vaccinations in larger outbreaks and later ones in smaller outbreaks, providing an alternative perspective to time-triggered campaign.

We explored counterfactual vaccination scenarios under various outbreak characteristics (**Table 2**). Vaccine coverage were set at 60% to 90% of the affected population—an administrative unit where the outbreak was reported. The vaccine coverages were based on trends observed in global OCV campaigns (2013-2018)⁶¹ and a recent campaign in

Ethiopia.⁶² We selected 75% coverage as a representative value for a single-dose regimen , reflecting coverage levels typically observed during OCV responses in humanitarian crises or outbreaks. This estimate also accounts for differences between estimated coverage and administrative coverage (Supplementary Information **S5**).

To account for the delay in developing protective immunity post-vaccination, we incorporated a period reflecting the time required for seroconversion—commonly defined as at least four-fold increase in vibriocidal antibody titers from baseline.^{40–48} Studies on OCV immunogenicity indicate that this increase can occur between 3 to 28 days (most often 7 to 14 days) following the first dose in a two-dose regimen.^{40,42–45} Accordingly we assumed the vaccine-derived protection would occur between 3 and 14 days (Supplementary Information **S6**).

Because vaccine supply is limited, not all outbreaks could be targeted for intervention. We therefore analyzed subsets of outbreaks that could have been covered with historically available vaccine supplies to estimate the potential overall impact on outbreaks throughout sub-Saharan Africa and to examine how various prioritization strategies might alter campaign outcomes. Specifically, we modeled the distribution of 185 million doses—the approximate number of single-dose OCVs deployed globally since the creation of the OCV stockpile through 2023—under three sampling strategies: random, attack rate-weighted, and cumulative cases-weighted sampling (**Table 2**; Supplementary Information **S7**). Random sampling treated that all outbreaks had the same probability of being targeted and served as a baseline, whereas weighting by attack rate or cumulative cases prioritized more severe or extensive outbreaks, reflecting scenarios where resources are allocated to settings with higher risk, potentially yielding a greater overall impact.

Table 2. OCV campaign settings

OCV campaign feature	Model settings that were tested
Campaign coverage	The entire population in which the outbreak was reported is targeted. Coverage varies from 60% to 90%, in increments of 5%.
Campaign triggers	Time-triggered OCV campaign: Vaccination begins 1 - 20 weeks after the outbreak starts, in increments of one week Case-triggered OCV campaign: Vaccination begins when the number of cumulative outbreak cases reached 1, 10, 20, 30, ..., 300, 400, ..., 1,000 cases.
Vaccination rounds	One round of vaccination per campaign.
Outbreak data	All outbreaks were analyzed both individually and collectively. Four subsets were created for collective analysis using different sampling strategies (uniform random, or attack rate -weighted or cumulative case-weighted.)

Cost-effectiveness analysis

We evaluated cost-effectiveness by calculating the cost per case, death, and DALY averted. Following a literature review (see Supplementary Material **S8**), we selected recent, high-quality cost estimates representative of our modeling data. To be comprehensive, we adopted a societal perspective that includes direct costs (e.g., vaccine procurement and healthcare utilization) and indirect costs (e.g., productivity losses). This perspective provides a holistic view of ORI interventions. All costs were adjusted to 2023 US\$ equivalents by inflating the costs from the original reporting year based on annual consumer price index changes for sub-Saharan Africa through 2023.^{63–65}

Results

The outbreaks

We identified 1,406 cholera outbreaks across sub-Saharan Africa between 2010 and 2023, excluding 258 OCV outbreaks where OCV interventions occurred. At the continental level Weekly incidence of cholera shows substantial temporal heterogeneity (**Figure 2A**).

Outbreaks vary substantially across size and duration with duration ranging from 5 to 139 weeks and outbreak sizes varying from 3 to 102,651 suspected cases (**Figure 2B**). The attack rate ranged from 0.002 to 6,542 rates per 10,000 people. Of the 152 outbreaks reporting at least one laboratory-confirmed case, duration ranged from 5 to 56 weeks, outbreak sizes ranged from 10 to 4,638 suspected cases, and attack rates were between 0.002 and 6,542 rates per 10,000 people. Geographically, the dataset covered much of the African continent, although no outbreaks were recorded in certain countries (**Figure 2C**). These patterns remained consistent across included and excluded outbreaks, suggesting that the dataset is broadly representative.

Impact of outbreak response immunization (ORI)

Time-triggered vaccination showed a sharp decline in percentage case reduction with increasing ORI delays. Across vaccine coverage levels of 60%, 75%, and 90%, the median reduction halved approximately every two weeks, decreasing by 31.6% per week (**Figure 3A**). By week 9, the median reduction fell below 5% for all coverage levels. Variability was pronounced with outliers showing close to 100% reduction while the median reduction being close to zero. Case-triggered vaccination exhibited similar trends, with the percentage case reduction decreasing as cumulative cases at the time of vaccination increased (**Figure 3B**). The median reduction dropped by 15.7% per 10 additional cases approximately and halved for every approximately 44 additional cases. The impact became negligible when vaccination began at more than 200 cumulative cases, regardless of coverage levels. Pre-emptive vaccination served as a benchmark, achieving reductions by 85.8% [95% PI: 59.3% - 100%], 94.8% [95% PI: 63.8% - 99.8%], and 98.9% [95% PI: 81.3% -

100%] at vaccine coverage levels of 60%, 75%, and 90%, respectively (maximum shown as dotted lines in **Figures 3A and Figure 3B**). Although ORI generally underperformed compared to pre-emptive strategies, some outlier outbreaks achieved comparable reductions even with delayed ORI. When stratifying by outbreak size, larger outbreaks consistently yielded higher median percentage case reductions for both time-triggered and case-triggered vaccination (**Figures 3C and 3D**). At a given vaccination scenario—a fixed week or a fixed cumulative case threshold—the case-triggered strategy demonstrated greater variability in percentage case reduction across outbreaks of different sizes.

The rapid decline in the percentage case reduction with delayed vaccination—whether measured by time (weeks) or cumulative cases—partly reflects growing number of outbreaks that had ended before vaccination began (**Figure S4A and S4B**). The importance of rapid intervention is further highlighted by weekly incidence trends aggregated across all outbreaks, which peak by week 3 and decline thereafter with 25% of all outbreak cases occurred by week 5 (**Figure S4C**). Case-triggered strategies exhibited greater variability in impact, indicated by a wider interquartile range (IQR) than time-triggered approaches (**Figure S4D**). This broader variability arises because a fixed cumulative-case threshold is reached at different times across outbreaks—earlier in larger outbreaks and later in smaller outbreaks.

Efficiency and cost effectiveness of the ORI

Table 3 summarizes the ORI impact under both trigger criteria at 75% vaccine coverage (see **Table S1** and **Table S2** for other coverage levels and campaign delay scenarios). In the time-triggered scenario, campaign delays from 1 to 15 weeks reduced the median case reductions from 70.9% to 0%, corresponding to 133 to 0 cases averted. The number of deaths averted decreased similarly. When campaigns began in the first week of an outbreak, the median number of cases averted was 0.6 per 1,000 doses; this figure gradually decreased with further campaign delays. The cost per case averted was approximately \$4,000 for a campaign initiated at week 1—below the cost-effectiveness

threshold of three times the GDP per capita of sub-Saharan Africa (approximately 1,636 USD⁵⁹), but increased substantially to \$22,000 if campaigns were delayed until week 13. For case-triggered campaigns, the overall patterns in cases and deaths averted resembled those of the time-triggered strategy, with substantial declines as the cumulative case count at vaccination increased. However, the number of cases averted per 1,000 doses remained relatively stable or varied less than under the time-triggered scenario, as did deaths averted per dose and cost per case, death, and DALY averted. This difference arises because the timing of vaccination differed across outbreaks; in large outbreaks, vaccination still occurred relatively early, yielding a high number of averted cases per dose, whereas in smaller outbreaks, vaccination took place late, producing minimal percentage case reduction and driving the median reduction to zero.

Cumulative impact, efficiency, and cost-effectiveness

ORI campaigns with 75% coverage initiated at week 9 demonstrated substantial impact when vaccine impacts were assessed collectively across sampled outbreaks (**Figure 4**). Over time, the cumulative number of cases averted by these campaigns increased reaching 27,649 [95% PI: 13,621 - 42,365] in 2022. The percentage reduction in cases were 17.8% [95% PI: 1.4 - 35.8] in 2013 and rose to 33.4% [95% PI: 19.2 - 45.3] in 2022, in contrast to 2.8% [95% PI: 0% - 74.3%] when the same vaccination scenario was applied to individual outbreaks (**Table 2**; **Figure 3**). This discrepancy may arise because large outbreaks, which typically see greater reductions (**Figure 3**), exert a disproportionate influence on overall estimates when impacts are aggregated. Across these sampled outbreaks, percentage case reduction decreased by 6.4% per week of delay in campaign initiation, halving every 11 weeks (**Figure S5**). In contrast, for individual outbreaks, the median weekly decline was approximately 30% (**Figure 3**). Case averted per 1,000 OCVs (**Figure 4D**) and the cost per case averted (**Figure 4E**) remained relatively stable over time, with variance decreasing as the number of deployed OCVs or the number of outbreaks increased. When assessed collectively across outbreaks, the estimated case averted per 1,000 OCVs were 0.63 [95% PI: 0.346 - 0.932], approximately fourfold higher than 0.141 [95% PI: 0 - 12.007] per 1,000

OCVs when assessed individually. Similarly, the cost per case averted were US\$5,491 [95%: 3,326 - 10,605] under the collective assessment, which was more than twofold lower than 14,167 (-444.8 - 1,087,944) when evaluated for individual outbreaks.

Discussion

Our study, encompassing 1,406 cholera outbreaks in sub-Saharan Africa from 2010 to 2023, demonstrates the substantial capacity of OCVs to mitigate the burden of cholera outbreaks. In particular, the timing and coverage of the ORI, and key outbreak characteristics—such as outbreak size, attack rate, outbreak duration—emerged as crucial determinants of overall impact and efficiency of OCVs. For a nine-week delay in initiating ORI (approximately the median delay observed in GTFCC dashboard), the median case reduction could fall below 5% in individual outbreaks, although substantial variability across outbreaks persists. When the results were aggregated across sampled outbreaks reflecting the number of OCV doses deployed during 2013-2022, however, ORIs still achieved a 33.4% [95% PI: 19.2 - 45.3] reduction in cases, primarily driven by their greater impact on larger outbreaks.

Our findings underscore the substantial potential of OCV campaigns at the aggregate level with some delay in campaign initiation. Furthermore, it implies the importance of prioritizing outbreaks to maximize impact. By selecting large outbreaks more often than would be selected uniformly at random, the ORI impact could be amplified further compared to uniformly randomized approach (**Table S3**). Although this prioritization approach assumes an idealized capacity to predict outbreak size at the time of a vaccination campaign, such prediction may be to some extent possible using known risk factors, including environmental indicators, socioeconomic conditions (e.g., poverty, literacy, healthcare access), and WASH infrastructure.^{19,66–69} These proxies can guide the distribution of limited OCV supplies to large or prolonged outbreaks. Targeting high-risk geographical areas has similarly been shown to bolster the impact and cost-effectiveness of OCV campaigns in sub-Saharan Africa.⁷⁰ Moreover, case-area targeted interventions during an outbreak have

demonstrated improved OCV efficiency.¹ As researchers continue to develop and validate predictive models, the ability to forecast outbreaks accurately—and thus inform more effective campaign decisions—will likely improve.

Many previous studies on OCV impact during or prior to cholera outbreaks have largely focused on single or a few large-scale events,^{1–4,6–8,10,16,71} in contrast to our extensive dataset (**Figure S7**). For example, pre-emptive vaccination with 50% coverage in Matlab, Bangladesh was shown to reduce cholera cases by 93% [95% CI: 82% - 99%].¹¹ This estimate slightly exceeds our predictions of 80.3% [95% PI: 53.8% - 97.4%]. This is, however, expected considering that the indirect effectiveness from the simulation model tended to be larger than those used in our model (**Figure S1**). Another modeling study of reactive, single-dose OCV campaigns in Zimbabwe, Haiti, and Guinea found respective case reductions of 71.6%, 65.3%, and 61.9%, if implemented on day 122, day 76, and day 61, respectively.¹⁵ Likewise, a single-dose OCV campaign in N'Djamena, Chad initiated on day 50 (or day 130 or day 180) reduced cases by 81% (or 43% or 8%).¹ Additional modeling of multiple strategies in N'Djamena, Maela (Thailand), and Haiti showed that a single-dose regimen for individuals over five years old often produced higher case reductions than other strategies, reducing up to about 68%, 90%, and 36% of total and symptomatic infections, respectively.¹⁴ Taken together, these studies highlight substantial impact of OCV campaigns, but their focus on larger outbreaks may inflate overall impact estimates. By contrast, our analysis of 1,406 outbreaks of varying sizes and attack rates indicates that while the impact of an OCV campaign can vary widely at the individual level, collectively, it still produces significant benefits.

Despite its strengths, our study has several limitations. First, the assumption of uniform vaccine coverage within administrative units may underestimate the potential benefits of more focused strategies, such as care-area targeted interventions^{1,72–74} or geographic targeting.⁷⁰ Conversely, vaccine wastage, as documented in coverage reports⁷⁵ could lead to overestimates of cases, deaths, and DALYs averted per 1,000 doses. Additional field data

would help balance these factors and refine future impact estimates. Second, our parameterization of indirect vaccine effectiveness are derived from Bangladesh-based studies,^{11,35} which may not fully generalize to sub-Saharan African contexts. Further research in diverse settings would improve robustness in these projections. Although further research is needed to strengthen these projections, the similarity between static and dynamic model outputs (**Figure S8**) suggests that our parameterization does capture indirect effects reasonably well. Our static modeling approach with indirect effectiveness from field observations may provide an alternative approach to modeling vaccine impact during outbreak response vaccination. Third, by focusing individual outbreaks (or aggregating them), our model assumes that the impact of vaccine is confined to a single ongoing outbreak. In reality, an ORI could serve as a pre-emptive measure for subsequent outbreaks in the same or neighboring areas. Due to a lack of geographic details for each outbreak and epidemiologic links between the outbreaks, we could not evaluate the downstream or spatial effects. Fourth, the reporting scale in our dataset could be an artifact in that multiple small outbreaks reported in administrative level 2 could actually be part of the same larger outbreak reported in administrative level 1. This may lead to underestimates of the vaccine impact as our results show that the impact of the vaccine greater for larger outbreaks. Fifth, our reliance on suspected rather than confirmed cases may overestimate the vaccine impact given that roughly half may not be true cholera.⁷⁶ However, underreporting of cholera in sub-Saharan Africa may partially offset this discrepancy and also analyses based on confirmed outbreaks only yielded similar results (**Tables S4 and S5**). Finally, our estimates are based on the outbreaks identified according to our definition, which requires at least three consecutive weeks of increasing cases above the outbreak threshold. Given the lack of a universally accepted definition of an outbreak, applying different criteria would produce different datasets and potentially different vaccine impact estimates. For example, using an alternative criterion of requiring at least seven weeks of increasing cases, our analysis produced broadly similar results, but with slightly higher percentage case reduction, more cases averted per vaccine dose, a slower decrease

in percentage case reduction with increasing delay in vaccination campaign, and greater cost-effectiveness (**Figure S9, Tables S6 and S7**).

Conclusion

This study provides key insights into the effectiveness of outbreak response immunizations (ORIs) across diverse cholera outbreak scenarios in sub-Saharan Africa. Our results highlight the critical role of timing and outbreak prioritization in maximizing vaccine impact, especially under resource constraints. Moving forward, refining predictive models, incorporating real-time surveillance data, and devising innovative vaccine distribution strategies will be essential. These efforts can enable more resilient, data-driven responses to cholera and help reduce the global burden of this disease.

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Figure legends

Figure 1. Overview of the modeling the impact of outbreak response immunization. (A) shows the process of the identification of suspected and confirmed cholera outbreaks and the selection of a subset for vaccine impact modeling. The outcome metrics are presented in the bottom box. (B) illustrates the hypothetical counterfactual outbreaks under a vaccination campaign. Weekly cases may decline after some delay after vaccination.

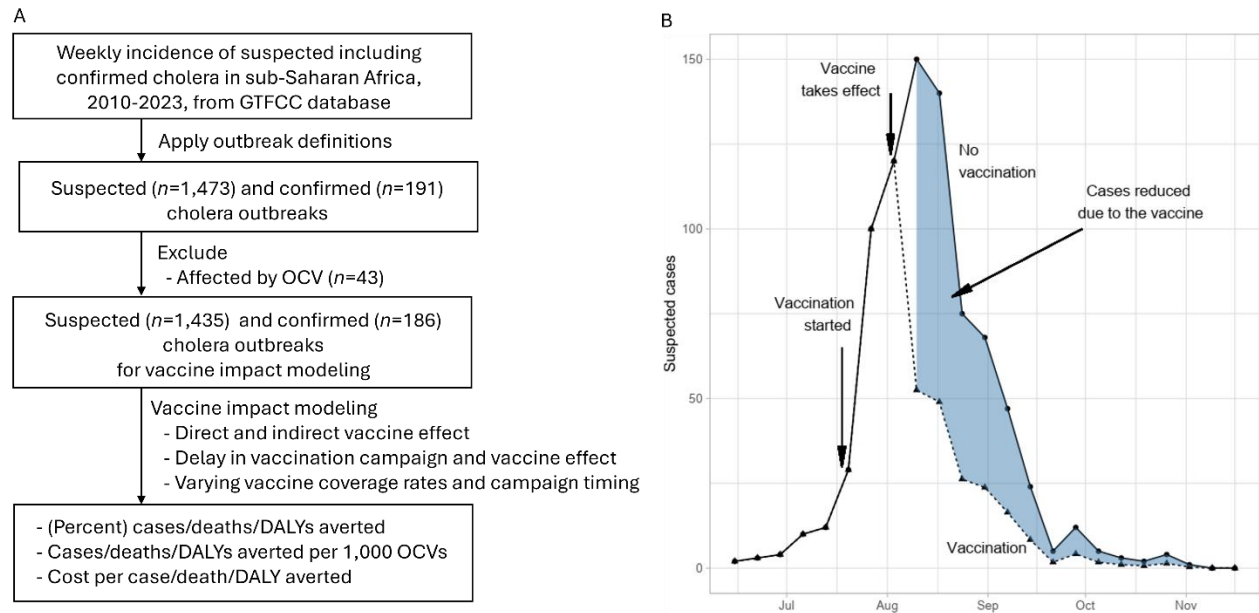


Figure 2. Outbreak data set in the study. Yellow, blue and red colors represent excluded, suspected, and confirmed cholera outbreaks, respectively. (A) weekly suspected cases across time. (B) displays the total number of suspected cases throughout the outbreak against the duration of the outbreak on a log scale for both X and Y axes. (C) shows the geographical location of the outbreak at the administrative unit levels 0,1, and 2.

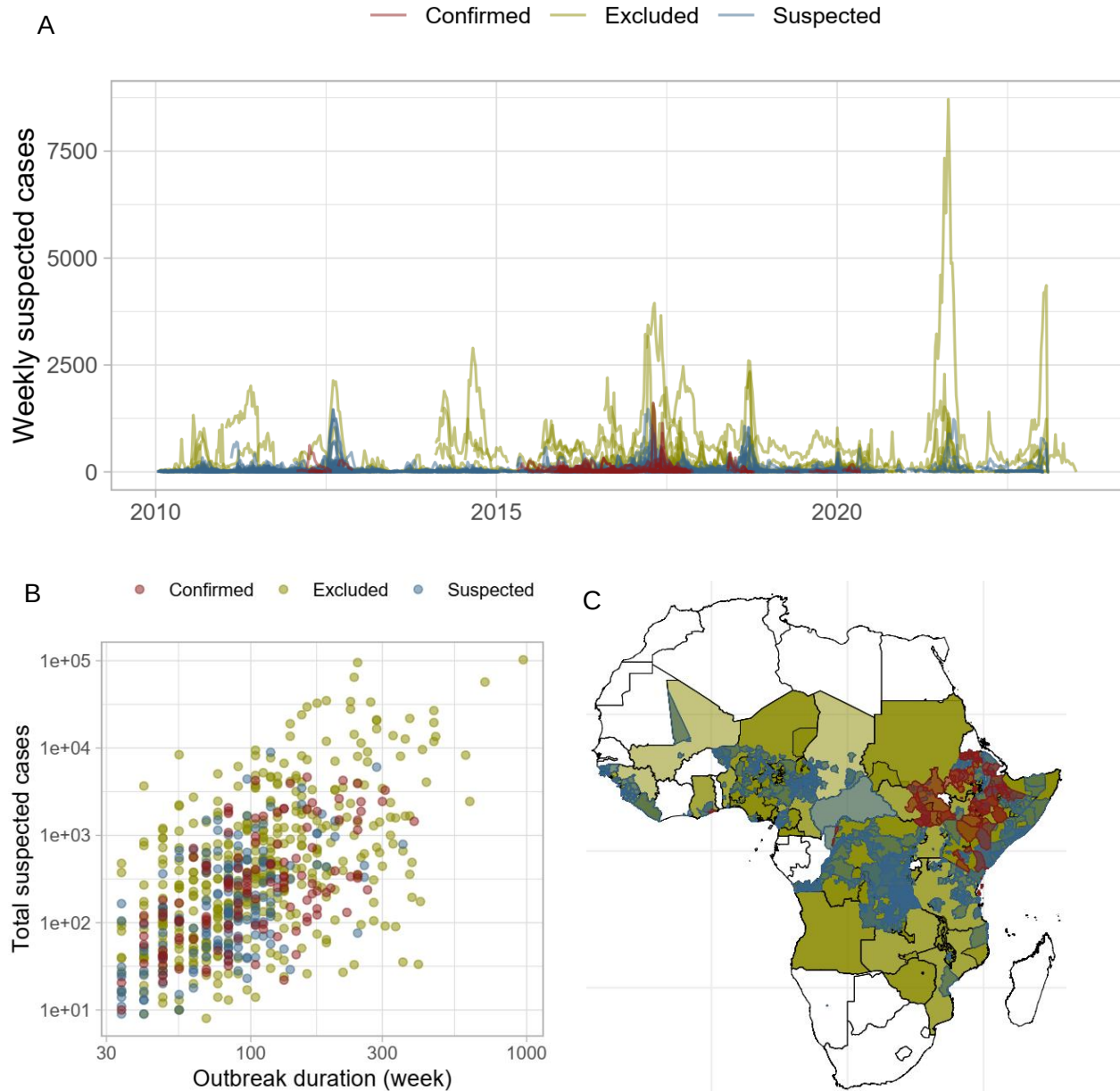


Figure. 3. Percent case reduction by the ORI during cholera outbreaks. **A** and **B** show the percentage case reduction at three vaccine coverage rates (i.e., 60%, 75%, and 90%) across time and case campaign triggers. Green, red, blue dotted lines that appear green show the maximum percentage case reduction that can be achieved via pre-emptive vaccination. The exponential decays (black lines) indicate the percentage case reduction of 31.6% by each week's delay (halving approximately 2.2 weeks) and 15.7% percent by 10 additional cases (having approximately every 44 additional cases). **C** and **D** show median case reduction (%) at 75% vaccine coverage across two campaign triggers for outbreak subsets categorized by the total number of cases for each outbreak. Y-axis label, $\geq P_{X\%}$, indicates a subset of outbreak data for which the total cases are larger than or equal to Xth percentile.

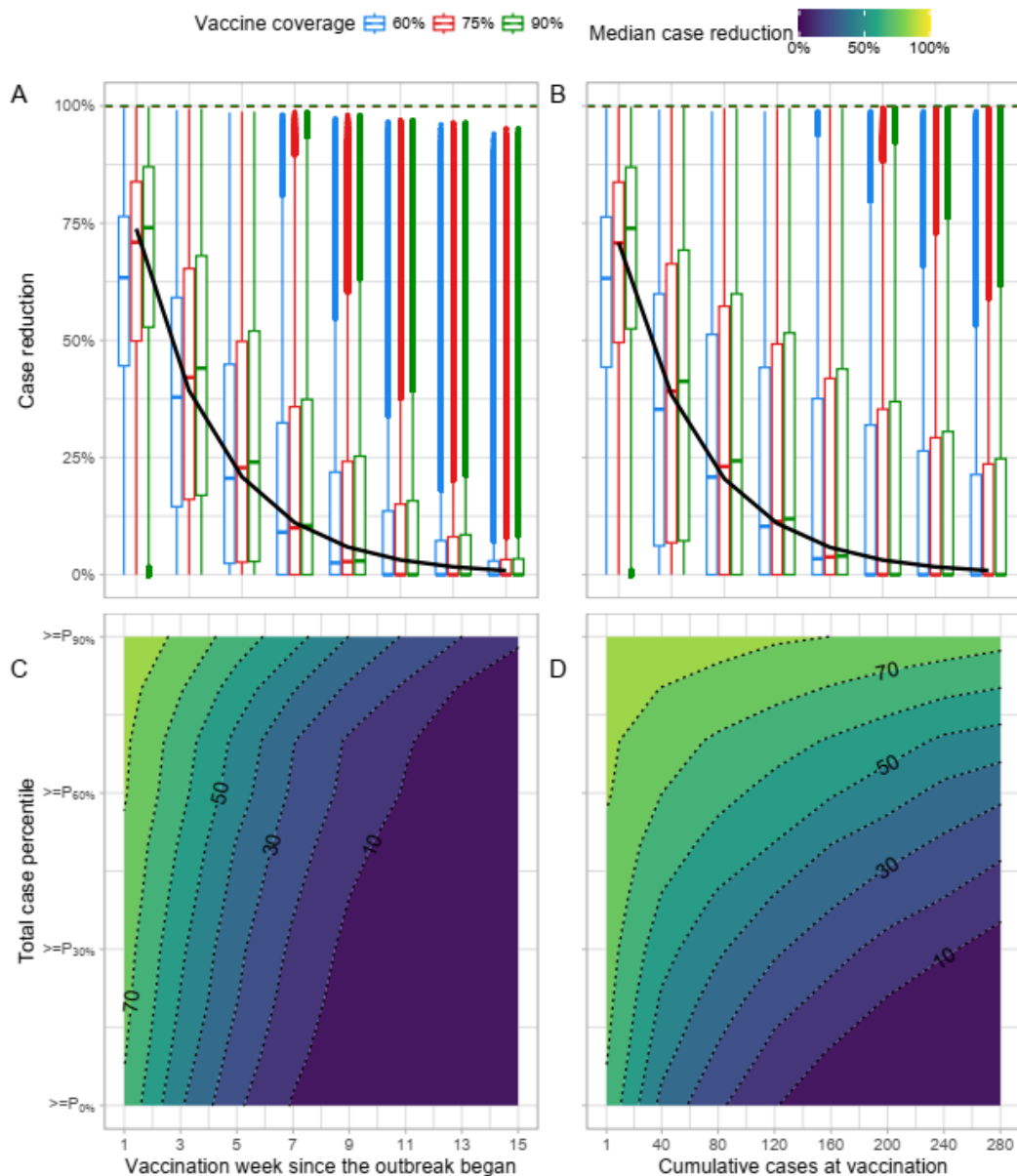


Figure. 4. Vaccine impact at week 9 under 75% coverage for subsets of outbreaks. The red dotted lines in panels (C), (D), and (E) indicate the median value across all outbreaks in the dataset when vaccine impact was assessed individually.

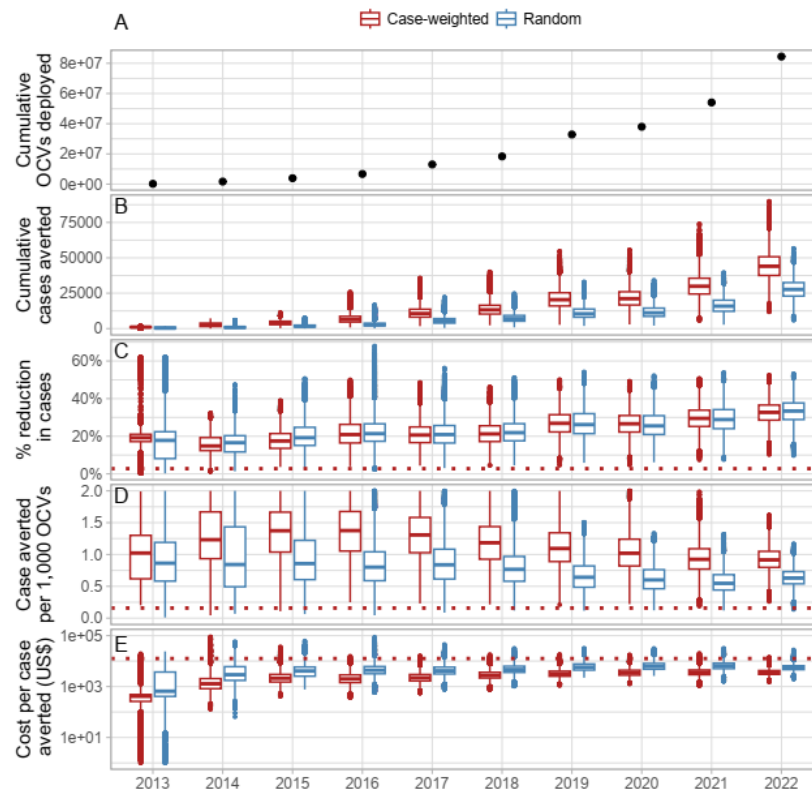


Table 1. Model parameters. (in the main text)

Table 2. OCV campaign settings (in main text)

Table 3. Impact of ORI with vaccine coverage 75%. Cases and deaths averted, OCV efficiency and costs in suspected cholera outbreaks observed in sub-Saharan Africa, 2010-2019. Values indicate the median with 95% prediction intervals.

Vaccination campaign triggers		% case reduction	Case averted	Death averted	Case averted per 1,000 OCVs	Death averted per 1,000 OCVs	Cost per case averted	Cost per death averted	Cost per DALY averted
Time-triggered (week)	1	70.9 (0 - 95.3)	133 (0 - 7,187)	1 (0 - 72)	0.594 (0 - 22.705)	0.004 (0 - 0.445)	5,532.7 (-1,459 - 482,034.2)	173,474.8 (-47,067.6 - 8,568,804)	34,062.5 (-1,722.3 - 311,871,248)
	5	22.8 (0 - 83.8)	39 (0 - 5,645)	0 (0 - 53)	0.231 (0 - 15.021)	0 (0 - 0.223)	10,693.7 (-673.2 - 1,065,526)	223,495.9 (-40,524.6 - 11,174,328)	359,013.9 (-1,057.2 - 801,282,144)
	9	2.8 (0 - 74.3)	5 (0 - 5,001)	0 (0 - 37)	0.141 (0 - 12.007)	0 (0 - 0.173)	14,167 (-444.8 - 1,087,944)	259,129.7 (-36,133.8 - 15,611,394)	1,162,673 (-868.4 - 818,128,084)
	13	0 (0 - 65.1)	0 (0 - 4,139)	0 (0 - 25)	0.107 (0 - 9.7)	0 (0 - 0.13)	18,002.9 (-285.4 - 1,506,539)	295,234.2 (-18,134.2 - 17,207,080)	2,384,529 (-398.2 - 1,072,901,744)
Case-triggered (case)	1	68.2 (22 - 88.5)	135 (5 - 7,050)	1 (0 - 73)	0.613 (0.009 - 22.747)	0.004 (0 - 0.449)	5,837 (-1,380.8 - 442,410.5)	226,802.5 (-43,155.5 - 23,041,844)	33,896.9 (-1,594.3 - 229,482,320)
	50	36.5 (0 - 86.8)	75 (0 - 7,050)	0 (0 - 67)	0.466 (0 - 19.828)	0.002 (0 - 0.273)	7,321.4 (-937.9 - 1,355,651)	277,135.3 (-43,837.5 - 25,139,306)	59,394.8 (-1,520.2 - 1,000,261,582)
	100	18.8 (0 - 85.5)	41 (0 - 7,046)	0 (0 - 67)	0.503 (0 - 22.885)	0.002 (0 - 0.321)	7,135.9 (-754.1 - 1,489,159)	285,477.3 (-42,109.9 - 30,825,782)	54,921.1 (-1,552.9 - 1,131,081,034)
	300	0 (0 - 81.7)	0 (0 - 6,571)	0 (0 - 56)	0.624 (0 - 17.421)	0.002 (0 - 0.218)	5,924.2 (-606.1 - 2,493,805)	317,056.4 (-40,698.9 - 36,497,610)	75,323.8 (-1,402.9 - 1,693,559,075)

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S1. Vaccine impact

Following the notations used in the previous study,¹ we define population A , in which a fraction, π , of the population is vaccinated and the rest, $1 - \pi$, of the population remains unvaccinated. We define another population, B , in which no one was vaccinated. Let C_{A0} and C_{A1} denote cumulative incidence in the unvaccinated and vaccinated individuals of population A . Similarly, let C_{B0} denote cumulative incidence in population B . Four measures of vaccine effectiveness—direct (DVE), indirect (IVE), total (TVE), and overall (OVE)—can be defined as follows:

$$\text{DVE} = 1 - \frac{C_{A1}}{C_{A0}}, \text{IVE} = 1 - \frac{C_{A0}}{C_{B0}}, \text{TVE} = 1 - \frac{C_{A1}}{C_{B0}}, \text{OVE} = 1 - \frac{\pi C_{A1} + (1 - \pi)C_{A0}}{C_{B0}}.$$

We see that the following relationships hold true.

$$\text{OVE} = 1 - [\pi(1 - \text{TVE}) + (1 - \pi)(1 - \text{IVE})],$$

$$\text{TVE} = 1 - (1 - \text{DVE})(1 - \text{IVE}),$$

In the static model, the impact of vaccine was modeled based on the definition of overall effectiveness, given the direct and indirect vaccine effectiveness, of which estimates are available. Following the notions based on the study,¹ the cumulative incidence, $C'_{i,\pi}$, for age group i in a population in which a proportion, π , of population was vaccinated can be defined as follows:

$$C'_{i,\pi} = C_i(1 - \text{OVE}).$$

where C_i , represents the incidence before vaccines are introduced in the population. We replace OVE in the equation using the DVE and IVE, for which estimates are available, based on the relationship we stated above:

$$C'_{i,\pi} = C_i[\pi(1 - \text{DVE})(1 - \text{IVE}) + (1 - \pi)(1 - \text{IVE})].$$

S2. Indirect vaccine effectiveness

Indirect vaccine effectiveness (IVE) of the oral cholera vaccine was estimated using incidence rates among unvaccinated individuals residing in clusters with varying vaccine coverage levels, as observed in a randomized trial conducted in Matlab, Bangladesh.² Vaccine coverage rates were reported as a range and the mid-point was taken as the representative value. Observed IVE at vaccine coverage x , was calculated as $1 - IR_x/IR_0$, where IR_x represents the incidence rate among unvaccinated people in a population with vaccine coverage rate x , and IR_0 denotes the incidence rate in a population with no vaccination. While IR_0 was not provided in the original analysis of the clinical trial, a subsequent dynamic modeling study³ provided a plausible value of IR_0 , which was used to compute IVE. This approach resulted in five IVE values at 5 vaccine coverage rates.

To model the relationship between IVE and vaccine coverage, we used a Bayesian Beta regression. The expected value of IVE, Y , was an expit-transformed function of η , which itself was modeled as a linear function of the independent variable, vaccine coverage rate X . The response variable Y was assumed to follow a Beta distribution, parameterized by its mean, μ , and precision, ϕ . Weekly informative priors were assigned to the linear predictor coefficients, β , and the precision parameter, $\phi > 0$, based on the standard recommendations for Bayesian analysis.⁴ The regression model was implemented in the probabilistic programming language Stan,⁵ and the code is available on GitHub.⁶

$$E(Y) = \mu = \frac{1}{1 + e^{-\eta}}$$

$$\eta = \beta_0 + \beta_1 X$$

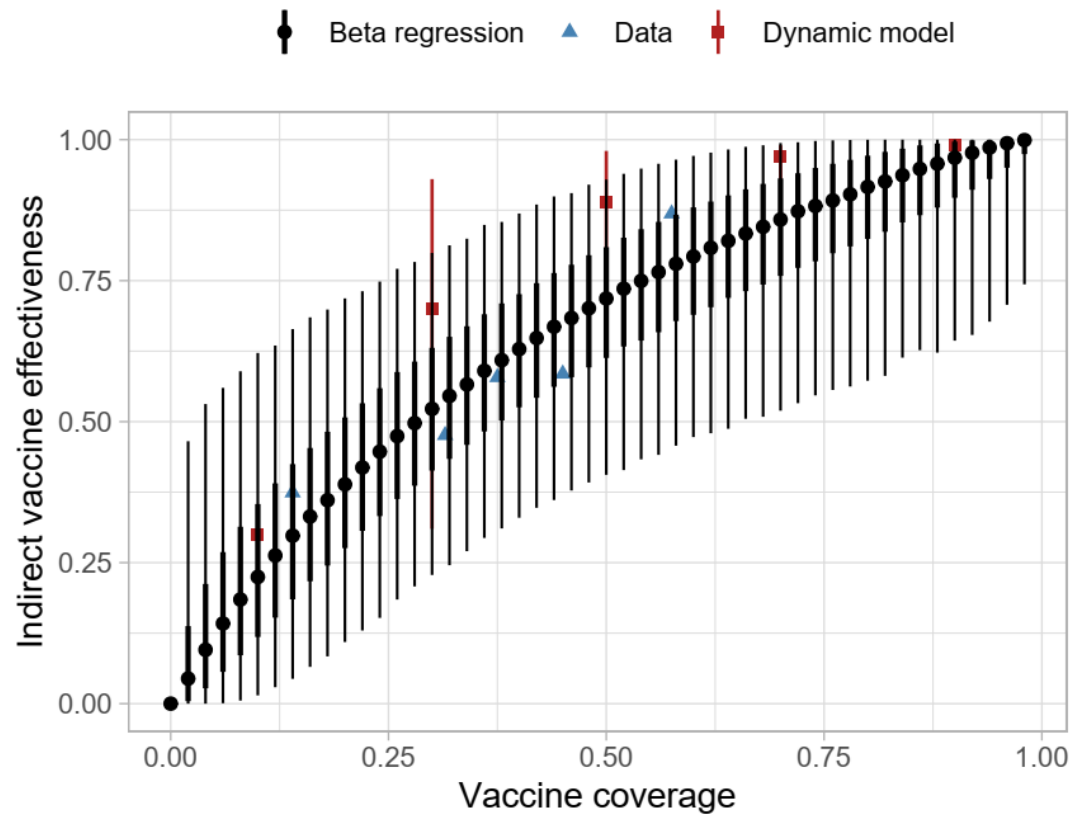
$$Y \sim \text{Beta}(\mu, \phi)$$

$$\beta \sim \text{Normal}(0, 5)$$

$$\phi \sim \text{Cauchy}(0, 5)$$

Figure S1.

Posterior predicted values for the indirect vaccine effectiveness based on a Bayesian beta regression with cholera incidence data from Matlab, Bangladesh. Dynamic model represents the values from the dynamic model by Longini et al.³



S3. Parameter exploration using Sobol low-discrepancy sampling

To comprehensively explore the effects of key model parameters on cholera outbreak dynamics and the impact of OCV campaigns, we conducted a systematic parameter sampling using the Sobol low-discrepancy sequence.^{7,8} This method provides an efficient way to thoroughly cover complex parameter spaces while minimizing clustering or overrepresentation of specific regions, making it particularly suitable for sensitivity and uncertainty analyses.

We examined a range of plausible values for the following parameters: vaccine efficacy (<5 years and 5+ years), indirect vaccine effectiveness, delay from vaccination and development of partially protective immunity, and campaign duration (days required to complete vaccination in a target population). Different distributions are assumed for each parameter based on the assumption of the estimation and on our best judgement. For the direct vaccine efficacy, we assumed that the log incidence rate ratio (e.g., vaccine efficacy = $1 - \text{incidence rate ratio}$) as the confidence interval of the original estimates may usually be constructed this way. Indirect vaccine efficacy was drawn from the posterior predictive values of the Bayesian beta regression model we have developed (**S2**). Delay from vaccination and development of partially protective immunity has been assumed to be uniformly distributed due to the paucity of the data. The duration of a vaccination campaign was assumed to follow a truncated normal distribution with the minimum and maximum based on the data and our judgement.

We employed the Sobol low-discrepancy sequence to generate pseudo-random numbers for each parameter, ensuring uniform coverage across their respective distributions. Each Sobol sequence element, ranging from 0 to 1, was transformed into the target parameter space using the inverse cumulative distribution function (CDF) of the specified distribution. The Sobol sequence generation was implemented using the ‘pomp’ R package, with custom routines to map values onto the respective distributions. (GitHub). A total of 200 parameter sets were utilized, covering the entire feasible range of values for each parameter.

Validation involved visually inspecting histograms of the generated parameter values to confirm that the transformations were accurate and that the values adhered to the intended distributions. Improved convergence was achieved by reducing variance in parameter estimates, which allowed for more precise sensitivity analyses with fewer iterations. Additionally, efficient coverage was ensured by adequately sampling all regions of the parameter space, thereby avoiding the clustering often seen with purely random sampling.

S4. Vaccine effectiveness comparison using dynamic models

To assess the robustness of the vaccine impact projections from our static modeling approach, we compared these predictions with those from a dynamic modeling approach. The dynamic model accounts for cholera transmissibility and population immunity, enabling it to capture outbreak-specific variations and feedback loops inherent to epidemic processes—such as the amplification of the force of infection with increasing cases or the mitigating effect of population immunity. While our primary analysis of the static model focused on reactive vaccination scenarios (i.e., outbreak response immunization, ORIs), the comparison between the static and dynamic models was conducted solely for pre-emptive vaccination scenarios. This decision was made after accounting for the limitation in coarse weekly time resolution of the static model and the dynamic model's limited ability to account for delays in vaccination rollout and subsequent immunological responses, which complicates a direct comparison for ORIs. Pre-emptive vaccination scenarios, on the other hand, provide a more straightforward alignment between the two approaches, allowing us to identify similarities and differences. These differences likely stem from the indirect vaccine effects, which are inherently captured in the dynamic model but must be externally specified in the static model.

We employed the standard *Susceptible-Exposed-Infected-Recovered* (SEIR) framework in which *E* and *I* states were modeled with two compartments such that the residence time in

each compartment follows an Erlang distribution with the shape of 2, rather than the unrealistic exponential distribution (**Figure S2**).

The following is a set of equations to represent the model. The first subscript, i , and the second subscripts, j , (both 1 or 2) represent the age groups (< 5 yo and 5+ yo) and the j th compartments for each state, respectively, except for the S state. S state and the vaccine efficacy, χ , have only one subscript representing the age group.

$$\lambda = \left\{ \beta \left(\sum_{i=1}^2 \sum_{j=1}^2 I_{i,j} + I_{i,j}^V + \zeta (A_{i,j} + A_{i,j}^V) \right) \right\} / N$$

$$\frac{dS_i}{dt} = -(\lambda_p + \lambda_w)S_i$$

$$\frac{dE_{i,1}}{dt} = \lambda S_1 - 2\epsilon E_{i,1}$$

$$\frac{dE_{i,2}}{dt} = 2\epsilon E_{i,1} - 2\epsilon E_{i,2}$$

$$\frac{dI_{1,1}}{dt} = 2(1 - \theta)\epsilon E_{i,2} - 2\gamma I_{i,1}$$

$$\frac{dI_{i,2}}{dt} = 2\gamma I_{i,1} - 2\gamma I_{i,2}$$

$$\frac{dA_{i,1}}{dt} = 2\theta\epsilon E_{i,2} - 2\gamma A_{i,1}$$

$$\frac{dA_{i,2}}{dt} = 2\gamma A_{i,1} - 2\gamma A_{i,2}$$

$$\frac{dR_i}{dt} = 2\gamma (I_{i,2} + A_{i,2})$$

$$\frac{dV_{i,1}}{dt} = vS_i - \lambda V_{i,1} - 2\delta V_{i,1}$$

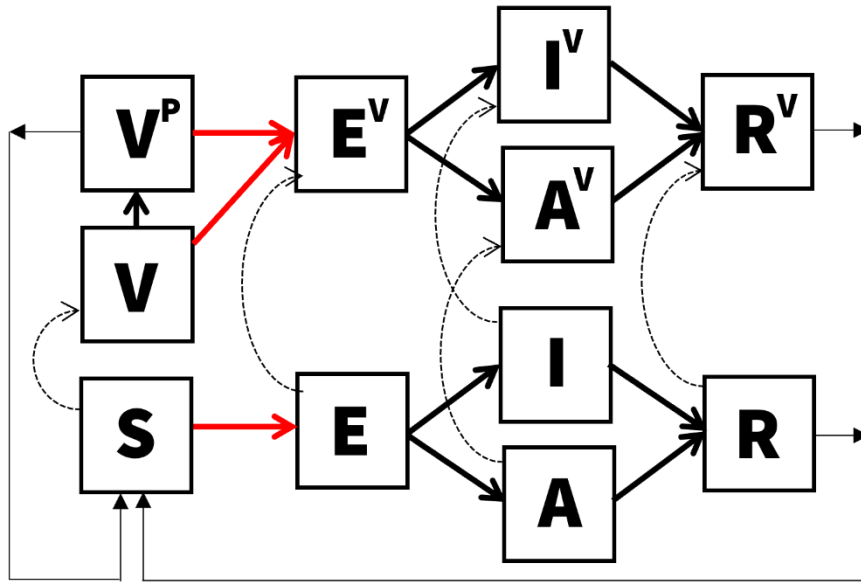
$$\frac{dV_{i,2}}{dt} = 2\delta (V_{i,1} - V_{i,2}) - \lambda V_{i,2}$$

$$\begin{aligned}
\frac{dV_{i,3}}{dt} &= 2\delta(V_{i,2} - V_{i,3}) - \lambda(1 - \chi_i)V_{i,3} \\
\frac{dV_{i,4}}{dt} &= 2\delta(V_{i,3} - V_{i,4}) - \lambda(1 - \chi_i)V_{i,4} \\
\frac{dE_{i,1}^V}{dt} &= \lambda(V_{i,1} + V_{i,2} + (1 - \chi_i)(V_{i,3} + V_{i,4})) - 2\epsilon E_{i,1}^V + vE_{i,1} \\
\frac{dE_{i,2}^V}{dt} &= 2\epsilon E_{i,1}^V - 2\epsilon E_{i,2}^V + vE_{i,2} \\
\frac{dI_{i,1}^V}{dt} &= 2(1 - \theta)\epsilon E_{i,2}^V - 2\gamma I_{i,1}^V \\
\frac{dI_{i,2}^V}{dt} &= 2\gamma I_{i,1}^V - 2\gamma I_{i,2}^V \\
\frac{dA_{i,1}^V}{dt} &= 2\theta\epsilon E_{i,2}^V - 2\gamma A_{i,1}^V + vA_{i,1} \\
\frac{dA_{i,2}^V}{dt} &= 2\gamma A_{i,1}^V - 2\gamma A_{i,2}^V + vA_{i,2} \\
\frac{dR_{i,1}^V}{dt} &= 2\gamma(I_{i,2}^V + A_{i,2}^V) + vR_{i,1} - 2\sigma R_{i,1}^V \\
\frac{dR_{i,2}^V}{dt} &= 2\sigma(R_{i,1}^V - R_{i,2}^V) + vR_{i,2}
\end{aligned}$$

Figure S2.

Compartmental flows for the dynamic model. S, E, I, A, R, V represent susceptible, exposed, infectious, asymptomatic, recovered, and vaccinated states. Superscript, P, for V represents those vaccinated and partially protected (rather than vaccinated but fully susceptible V state). Superscript, V, for E, I, A, and R represents vaccinated but infected population. Each compartment is modeled with two sub-compartments representing age

groups (<5 yo or 5+ yo) and E, I, A, R, V, and Vp states are additionally modeled with two compartments to model the Erlang-distributed residence times.



The incubation period was modeled using an Erlang distribution. This was achieved by fitting the Erlang distribution to a dataset of 1,000 samples, which were generated through a log-normal distribution with a median of 1.4 days and a dispersion of 1.98, as determined by a systematic literature review.¹ In our approach, we varied the shape parameter of the Erlang distribution by setting it to 1, 2, 3, and 4. These values represent the potential number of compartments for the incubation state within the ODE model of cholera transmission. The rate parameter was the only variable allowed to be adjusted and was determined through the maximization of the log likelihood. The selection of the optimal shape parameter was based on the comparison of likelihood values, ultimately identifying a shape of 2 and a rate of 1.115044 as the most suitable for our model.

We tried fitting the model to weekly incidence for all 1,406 outbreaks by calibrating four parameters: the proportion of the population that is susceptible in the beginning of the outbreak, s_0 , basic reproduction ratio, R_0 , initial proportion of people already infected at the start of the outbreak, i_0 , and the proportion of effective population size, n_0 . The n_0 parameter was introduced to address the possibility that the population experiencing the outbreak is likely to have been a fraction of the population of the administrative unit where

the outbreak was reported. Fitting involved assuming that the number of observed cases at a given week, $y(t)$, follows a Poisson distribution with its mean, $Y(t)$, and modelled incidence of symptomatic cases. The likelihood function $\mathcal{L}(\theta)$ of the model, defined by the parameters θ (as shown below), was maximized using the differential evolution method.⁹

$$L(\theta; y_1, \dots, y_n) = \prod_{j=1}^n \frac{e^{-Y_j} Y_j^{y_j}}{y_j!}$$

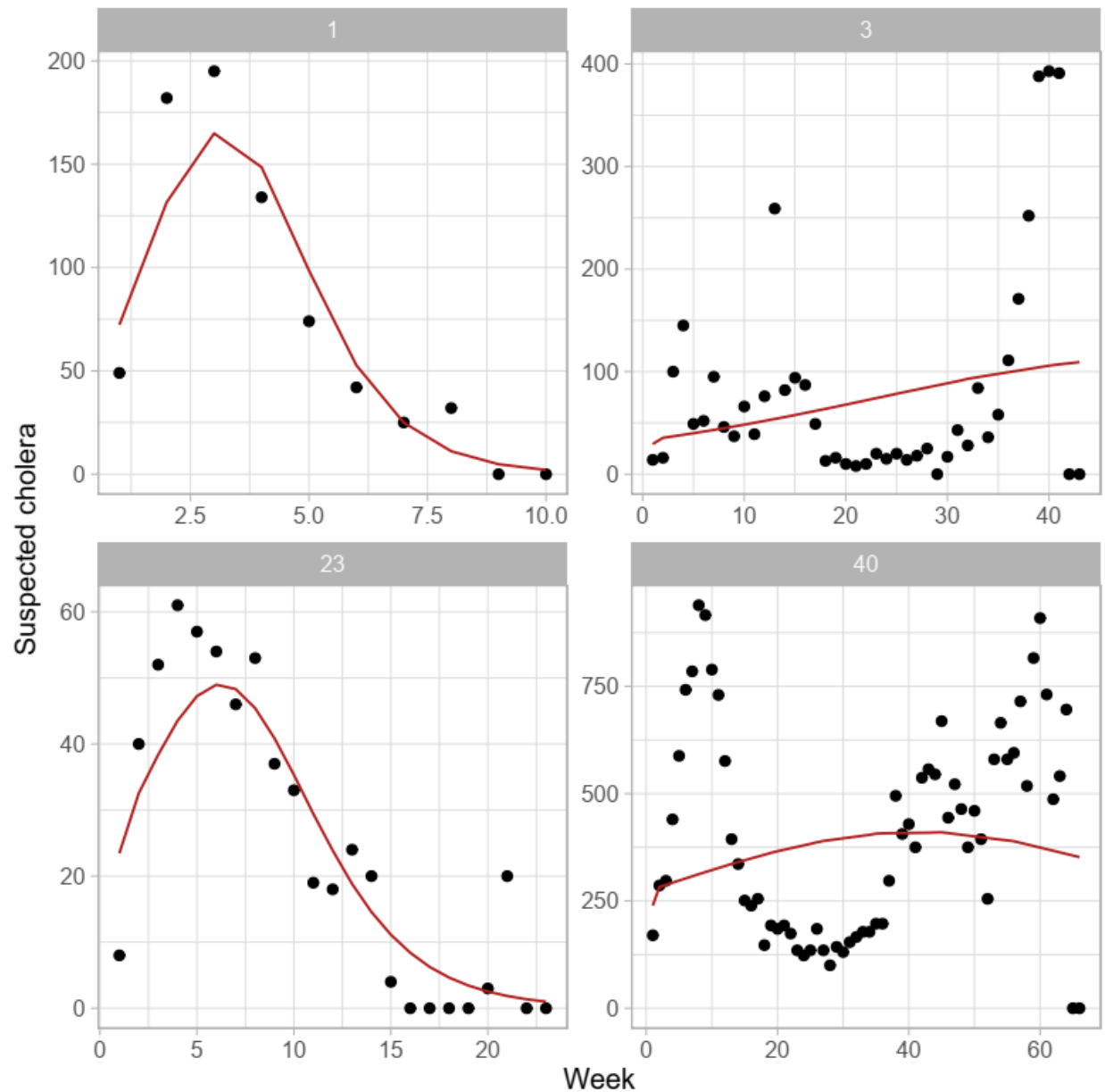
The differential evolution algorithm is a stochastic search algorithm and uses Latin Hypercube sampling as initial parameter values to explore within the bounds and local maxima. We explored a wide range of bounds for parameters to make sure that all reasonable values are examined. Acknowledging that the algorithm may end up in a local minimum by chance, we ran the algorithm with five different random seeds and chose the values that maximized the log likelihood that the algorithm reached at least twice.

After reviewing the fitting results, we selected 341 outbreaks where the model captured the dynamics of the outbreak sufficiently well by excluding fits where a) modeled and observed weekly incidences diverged upon visual inspection, b) parameter estimates were too close (0.1%) to the bounds of parameter inputs, raising concerns that the estimated parameter values were not robust, and c) the number of data points was smaller than 8, corresponding to twice the number of data points. While the number of parameters to be estimated is four, we consider the possibility that the number of data points may need to be much larger to inform the parameter values in practice. The distribution of the parameter estimates for the 341 outbreaks The dynamic model was implemented in Julia language¹⁰ and codes used are available on GitHub.

Figure S3.

Comparison of weekly incidence between the dynamic model and the observed data. Numbers at the top indicate IDs assigned to each outbreak. For outbreaks with IDs 1 and 23, the dynamic model (red lines) successfully captured the observed data (black dots).

Conversely, outbreaks with IDs 3 and 40 exhibited complex patterns that the dynamic model failed to accurately represent.



S5. Vaccine coverage rates for the single-dose regimen

We explored the vaccine coverage rates ranging from 60 to 95 percent of the population.

Vaccine coverage rates were chosen to account for observations from OCV campaigns

during 2013-2018¹¹ and a recent OCV campaigns in Ethiopia.¹² In particular, we chose 75% as the most representative value for a single-dose regimen. This value was obtained by first computing the ratio of estimated coverage to the administrative coverage, r , and then multiplying r with the observed administrative coverage rates during the first dose, AC_1 . r was computed by taking the ratio of estimated coverage to administrative coverage for the at least one dose from the OCV campaigns. ($EC_{\geq 1}$ and $AC_{\geq 1}$, respectively). $EC_{\geq 1}$ were available in the study and were derived assuming random allocation during vaccination during the first and the second dose. Let AC_1 and AC_2 be administrative vaccine coverage rates for the first and the second dose, respectively. Then,

$$AC_{\geq 1} = 1 - (1 - AC_1)(1 - AC_2)$$

$$r = \frac{EC_{\geq 1}}{AC_{\geq 1}}$$

In particular, we used $EC_{\geq 1}$ during humanitarian crises or outbreaks (73.2%, and 86.3%, respectively). Finally, we computed the estimated coverage rates for the first dose during humanitarian crises or outbreaks (62.6% and 85.3%) using the following relationship.

$$EC_1 = rAC_1$$

Arithmetic average of EC_1 during humanitarian crises or outbreaks is 74% and used 75% as a convenient proxy value.

S6. Immunogenicity of oral cholera vaccine

We assumed a period of delay before protective immunity develops following vaccination. Studies investigating the immunogenicity of OCV typically measure the rate of seroconversion (considered a proxy for protection), defined as a four-fold or greater increase in antibody/vibriocidal titres compared to baseline titres^{13–21}. However, dosing schedules, follow-up times, and populations varied across studies. For instance, Ng'ombe et al. investigated a two-dose schedule for the Shanchol OCV in Zambia, administering the

first and second doses 28 days apart and collecting blood samples at baseline and after day 28 (before second dose), month 6, month 12, month 24, month 30, month 42, and month 48¹³. They observed a significant rise in vibriocidal antibody titres at day 28, which declined nearly to baseline levels at month 6. Other studies by Mohan et al. (India) and Chowdhury et al. (Bangladesh) investigated a two-dose schedule for Shanchol with doses being administered 14 days apart instead, and blood collections performed at days 0, 14 (before second dose), and 28^{15,16}. Both studies found that vibriocidal antibody titres were significantly increased at day 14 and day 28, compared to baseline, and that this increase was higher after the first dose, compared to the increase following the second dose of vaccine. Charles et al. (Haiti) investigated a two-dose Shanchol schedule administered 28 days apart, with blood collection occurring 7 days post-vaccination for each dose¹⁷. They found that each age cohort had a significant vibriocidal antibody response by as early as day 7, compared to baseline, and increased further by day 21. Additionally, O-antigen specific IgA serum responses were also significant at day 7 and 21. We believe that these findings make it reasonable to conclude a time to immunity post-vaccination of 14 days for modeling purposes. This assumption will be tested via sensitivity analyses using a range of values (from 7 to 28 days) to explore the robustness of our results.

S7. Outbreak data subsets

The outbreak dataset is extensive and not all outbreaks may be targeted with OCVs given resource constraints. To prioritize outbreaks for vaccination, we created a subset of outbreaks that approximately 84 million doses of OCV equivalent to the number of OCVs deployed from 2013 to 2022, may be targeted, assuming a single-dose regimen. We developed random and case-weighted sampling approaches to select outbreaks, each designed to capture different strategic priorities in cholera control. Random sampling approach serves as a baseline approach where each outbreak has an equal probability of selection. It provides an unbiased representation of outbreaks regardless of their characteristics and would be useful for understanding the general distribution of outbreak characteristics in the subset. For the weighted sampling methods, the probability of i th observation is being selected, p_i , is determined as follows:

$$p_i = \frac{X_i}{\sum_{j=1}^n X_j},$$

where X_i could be cumulative number of cases of the i th outbreak. Case-weighted approach prioritizes outbreaks with larger total case counts where vaccination could prevent the greatest absolute number of cases. The method provides flexibility in outbreak selection based on different public health priorities while ensuring that the selected outbreaks remain within the constraints of available vaccine supplies. This approach can be adapted as vaccine availability changes or as different aspects of outbreak characteristics become priorities for intervention.

S8. Cost-effective analyses (CEA)

These analyses require input parameter values, including vaccine purchasing price, vaccine delivery cost, and costs related to the disease itself. Several studies reporting on these costs related to cholera can be found in the literature, however, differences in contexts and costing methods result in varying values. For the purposes of this CEA, a comprehensive literature review was conducted and values (and corresponding uncertainty ranges) were selected based on study recency, generalizability to our modeling data (which is primarily from African countries), and quality. Firstly, purchasing price of cholera vaccine will depend on the type of OCV. Currently, there are two WHO-prequalified cholera vaccines that are available for mass vaccination campaigns through the Global OCV Stockpile: Shanchol (US\$ 1.85) and Euvichol (US\$1.20) ^{22–24}. Secondly, there are no current studies that investigate the delivery costs for Euvichol, and few studies that do so for Shanchol, which provide a range between US\$1.13 to US\$3.60 ^{22,25–27}. Our analyses used a cost of delivery of \$1.48 from Teshome et al. since the study occurred in an African country (Ethiopia) and it provides a practical midpoint ²⁶. Thirdly, there are a number of studies that report the cost of illness due to cholera. These typically include direct out-of-pocket and health care system costs, and indirect costs associated with lost productivity. For these analyses, we decided to utilize inputs reported by Mogasale et al. ²⁸. This 2021

paper performed a review of the available literature on cholera cost of illness in African countries and reported the mean values (and minimum and maximum values) for out-of-pocket costs for hospitalised (US\$38.87, range US\$6.05 to US\$109.94) and outpatient cases (US\$5.71, range US\$0.89 to US\$16.15), public health service delivery costs for hospitalised (US\$65.77, range US\$39.83 to US\$85.41) and outpatient cases (US\$2.84, range \$2.55 to \$3.31), and number of days with loss of income for cholera cases (5.76 days, range 4.61 to 6.97) and caregivers (3.86 days, range 3.06 to 4.64) across these studies ²⁸. Furthermore, discount rate was set to 3%, length of illness was set to 4 days, and disability weight was set to 0.2, based on previous studies ^{22,29–31}. To account for the spectrum of cholera severity in the analysis, length of illness in severe state was set to 0.02 years and disability weights were stratified into mild (0.074, 95%CI: 0.049-0.104), moderate (0.188, 95%CI: 0.125-0.264), and severe cases (0.247, 95%CI: 0.164-0.348) ^{31,32}. The distribution of cholera severity was obtained from a seroepidemiologic study by Jackson et al., where, of participants with vibriocidal titers ≥ 3320 , 74.5% were asymptomatic, 6.9% had mild disease, 9.7% had moderate disease, and 9.0% had severe disease ³³.

Disability-adjusted life years (DALYs) were computed as:

$$\text{DALYs} = \text{Years of life lost from mortality (YLLs)} \\ + \text{Years of life lost to due to disability (YLDs)}$$

YLLs were computed by multiplying the number of deaths with life-expectancy at the average age of death. Average age of death was derived from the average of infections observed during outbreaks and clinical trials. Life expectancy at a given age, mean remaining years of life expected by an individual at that age, was derived from United Nations Life Expectancy at Exact Age for both sexes.³⁴ YLDs were computed by multiplying the number of cases with the duration of the disease in years and the disability weight. The duration of disease was assumed to be 2 days and the disability weight was assumed to be 0.216 to account for the moderate and severe cases a based on the disability weights from the 2019 Global Burden of Disease Study,³² assuming that reported cases in our data set were moderate or severe cases in our data set.

We evaluated cost-effectiveness by calculating the cost per case, death, and DALY averted. Costs related to the cholera vaccine and illness were selected based on recency, generalizability to our modeling data, and quality after a literature review. Prices were adjusted to 2023 USD\$ cost-equivalents by converting the costs from the reporting year and inflating them based on the sub-Saharan African average annual consumer price index percent change for each year up until 2023.^{35–37} Consumer prices were obtained from the publicly available International Monetary Fund database.³⁸ This was performed using the following equation:

$$\text{Cost}_{2023} = \text{Cost}_t \times \prod_{t+1}^{2023} \left(1 + \frac{\text{CPI change}_t}{100} \right),$$

where t is the year that the original cost was reported, and CPI change_t is the consumer price index annual percent change for each year t .

The vaccine procurement cost (\$1.65) and shipping cost (\$0.06) were sourced from UNICEF reports published in 2024, which did not specify the USD year.^{39,40} We assumed these costs were comparable to 2023 and, therefore, left them unadjusted. After adjusting for inflation, cost of vaccine delivery was set to \$2.32. Out-of-pocket costs for those hospitalized was set to \$109.32 and out-of-pocket costs for outpatient cases was set to \$16.06. Costs on the public health system for those hospitalized and outpatient cases was set to \$184.97 and \$7.99, respectively. Other parameters, such as number of days with loss of income for cholera cases and caregivers, were also obtained from the literature and are 5.76 and 3.86 days in the analysis, respectively²⁸. Length of illness was set to 4 days. To account for the spectrum of cholera severity, disability weights were stratified into mild (0.074, 95%CI: 0.049-0.104), moderate (0.188, 95%CI: 0.125-0.264), and severe cases

(0.247, 95%CI: 0.164-0.348)^{31,32}, with a population distribution of 74.5% asymptomatic cases, 6.9% mild, 9.7% moderate, and 9.0% severe cases³³.

Figure S4.

Outbreaks ending before the delayed vaccination campaign and outbreak size distribution across vaccination week and cumulative cases at vaccination. **A** and **B** show outbreaks (%) that are missed by delayed campaign or in addition by delay in vaccine taking effect. **C** and **D** show the distribution of suspected cases for each week and the distribution of weeks for each cumulative case, respectively. Left (**A** and **C**) and right (**B** and **D**) columns show results from time-triggered and case-triggered vaccination scenarios, respectively. The solid line in **C** represents weekly cases summed across outbreaks.

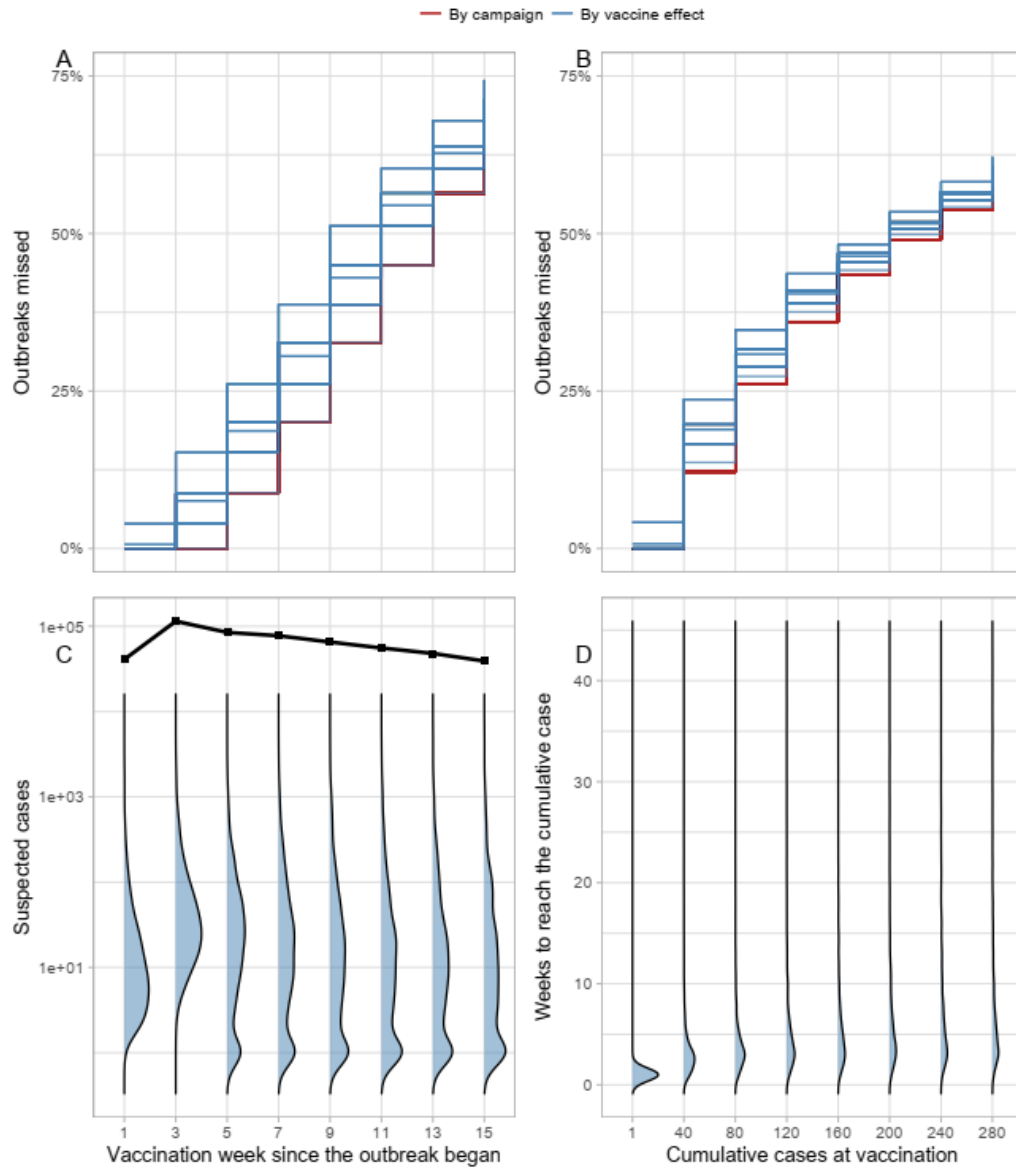


Table S1.

Cases and deaths averted, OCV efficiency and costs in suspected cholera outbreaks observed in sub-Saharan Africa, 2010-2022 for time-triggered vaccination scenarios. Metrics were assessed for individual outbreaks.

Campaign week	Vaccine coverage	% case reduction	Case averted	Death averted	Case averted per 1,000 OCVs	Death averted per 1,000 OCVs	Cost per case averted	Cost per death averted	Cost per DALY averted
1	0.6	z	120 (0 - 6,522)	1 (0 - 65)	0.675 (0 - 26.16)	0.004 (0 - 0.504)	4,807.7 (-1,675.1 - 423,710.5)	147,902.4 (-50,583.1 - 7,539,054)	30,066.3 (-1,868.1 - 276,272,034)
5	0.6	20.6 (0 - 78)	35 (0 - 5,184)	0 (0 - 48)	0.262 (0 - 17.233)	0 (0 - 0.254)	9,325.6 (-762.4 - 947,421.3)	192,818.3 (-44,238.5 - 9,972,158)	308,701 (-1,216.7 - 710,170,711)
9	0.6	2.5 (0 - 68.6)	4 (0 - 4,548)	0 (0 - 33)	0.16 (0 - 13.794)	0 (0 - 0.195)	12,307.3 (-518.5 - 968,777.1)	223,015.2 (-40,153.7 - 13,781,393)	1,016,493 (-1,099.3 - 744,327,590)
13	0.6	0 (0 - 59.8)	0 (0 - 3,664)	0 (0 - 23)	0.121 (0 - 11.019)	0 (0 - 0.152)	15,941.4 (-379.6 - 1,342,322)	255,466.3 (-24,148.7 - 15,274,743)	2,080,830 (-577.5 - 964,149,939)
17	0.6	0 (0 - 50)	0 (0 - 2,301)	0 (0 - 13)	0.13 (0 - 7.921)	0 (0 - 0.144)	15,067.5 (-285 - 1,389,936)	263,129 (-23,565.5 - 19,257,356)	1,606,477 (-294.2 - 1,081,417,804)
1	0.75	70.9 (0 - 95.3)	133 (0 - 7,187)	1 (0 - 72)	0.594 (0 - 22.705)	0.004 (0 - 0.445)	5,532.7 (-1,459 - 482,034.2)	173,474.8 (-47,067.6 - 8,568,804)	34,062.5 (-1,722.3 - 311,871,248)
5	0.75	22.8 (0 - 83.8)	39 (0 - 5,645)	0 (0 - 53)	0.231 (0 - 15.021)	0 (0 - 0.223)	10,693.7 (-673.2 - 1,065,526)	223,495.9 (-40,524.6 - 11,174,328)	359,013.9 (-1,057.2 - 801,282,144)

9	0.75	2.8 (0 - 74.3)	5 (0 - 5,001)	0 (0 - 37)	0.141 (0 - 12.007)	0 (0 - 0.173)	14,167 (-444.8 - 1,087,944)	259,129.7 (-36,133.8 - 15,611,394)	1,162,673 (-868.4 - 818,128,084)
13	0.75	0 (0 - 65.1)	0 (0 - 4,139)	0 (0 - 25)	0.107 (0 - 9.7)	0 (0 - 0.13)	18,002.9 (-285.4 - 1,506,539)	295,234.2 (-18,134.2 - 17,207,080)	2,384,529 (-398.2 - 1,072,901,744)
17	0.75	0 (0 - 55.2)	0 (0 - 2,532)	0 (0 - 14)	0.114 (0 - 6.958)	0 (0 - 0.124)	17,131.6 (-126.5 - 1,592,361)	304,090.3 (-15,524.9 - 23,296,291)	1,806,850 (-171 - 1,227,312,224)
1	0.9	74 (0 - 97.1)	139 (0 - 7,530)	1 (0 - 76)	0.516 (0 - 19.653)	0.003 (0 - 0.389)	6,449.6 (-1,243.5 - 546,603.7)	204,038.9 (-42,814.4 - 9,840,415)	39,494.5 (-1,459.4 - 356,400,857)
5	0.9	24 (0 - 86.5)	41 (0 - 5,944)	0 (0 - 56)	0.2 (0 - 12.906)	0 (0 - 0.193)	12,381.6 (-494.6 - 1,235,384)	260,551.3 (-37,629.6 - 12,984,447)	417,962 (-987.8 - 917,351,541)
9	0.9	2.9 (0 - 76.8)	5 (0 - 5,183)	0 (0 - 38)	0.122 (0 - 10.355)	0 (0 - 0.149)	16,445.1 (-351.4 - 1,254,116)	304,272.2 (-30,946.7 - 18,139,001)	1,363,377 (-727 - 916,967,958)
13	0.9	0 (0 - 67.5)	0 (0 - 4,383)	0 (0 - 26)	0.093 (0 - 8.387)	0 (0 - 0.113)	20,645.5 (-99.6 - 1,719,769)	344,613.8 (-15,980.8 - 19,841,520)	2,758,140 (-173.9 - 1,216,690,912)
17	0.9	0 (0 - 57.7)	0 (0 - 2,636)	0 (0 - 15)	0.1 (0 - 6.087)	0 (0 - 0.11)	19,789.9 (-40.2 - 1,818,713)	358,109.2 (-14,885.2 - 26,752,721)	2,061,254 (-38.3 - 1,417,275,879)

Table S2.

Cases and deaths averted, OCV efficiency and costs in suspected cholera outbreaks observed in sub-Saharan Africa, 2010-2022 for case-triggered vaccination scenarios. Metrics were assessed for individual outbreaks.

Case-trigger	Vaccine coverage	% case reduction	Case averted	Death averted	Case averted per 1,000 OCVs	Death averted per 1,000 OCVs	Cost per case averted	Cost per death averted	Cost per DALY averted
1	0.6	63.2 (0 - 92.2)	120 (0 - 6,522)	1 (0 - 65)	0.674 (0 - 26.16)	0.004 (0 - 0.504)	4,790.3 (-1,675.1 - 424,819.7)	147,866.5 (-50,622.5 - 7,551,919)	30,088.6 (-1,869.6 - 277,186,189)
50	0.6	31.1 (0 - 85)	63 (0 - 6,487)	0 (0 - 63)	0.523 (0 - 23.38)	0.001 (0 - 0.343)	5,512.5 (-1,256.6 - 442,108.8)	169,803 (-50,273.5 - 9,166,378)	48,401.3 (-1,833.2 - 316,058,919)
100	0.6	15 (0 - 82.6)	33 (0 - 6,376)	0 (0 - 61)	0.567 (0 - 26.003)	0.001 (0 - 0.354)	5,115.4 (-1,160.8 - 517,716.1)	179,492.8 (-48,839.8 - 10,888,498)	46,747.9 (-1,833.6 - 357,592,827)
500	0.6	0 (0 - 72.9)	0 (0 - 5,882)	0 (0 - 49)	0.655 (0 - 18.014)	0.002 (0 - 0.186)	4,658.5 (-815.9 - 555,116.7)	239,965 (-36,420.9 - 14,476,470)	45,262.4 (-1,455.6 - 391,228,061)
1000	0.6	0 (0 - 65.3)	0 (0 - 5,320)	0 (0 - 39)	0.638 (0 - 16.094)	0.002 (0 - 0.207)	5,593 (-181.4 - 434,839.8)	454,129 (-28,642.3 - 33,579,264)	61,749.1 (-747.1 - 278,576,941)
1	0.75	70.8 (0 - 95.2)	133 (0 - 7,187)	1 (0 - 72)	0.593 (0 - 22.705)	0.003 (0 - 0.445)	5,512.3 (-1,461.4 - 482,731.1)	173,443.2 (-47,174.6 - 8,600,315)	34,076 (-1,723.1 - 312,758,539)
50	0.75	34.5 (0 - 90.8)	70 (0 - 7,148)	0 (0 - 69)	0.464 (0 - 20.402)	0.001 (0 - 0.3)	6,297.1 (-1,044 - 496,514.2)	197,911.9 (-47,168.6 - 10,368,709)	55,400.9 (-1,666.6 - 361,723,892)
100	0.75	16.6 (0 - 88.8)	36 (0 - 7,014)	0 (0 - 68)	0.501 (0 - 23.155)	0.001 (0 - 0.306)	5,830.7 (-925.6 - 585,943)	207,570.8 (-45,897.3 - 12,128,792)	53,101.7 (-1,709.6 - 402,519,304)
500	0.75	0 (0 - 79.8)	0 (0 - 6,443)	0 (0 - 55)	0.574 (0 - 15.905)	0.002 (0 - 0.163)	5,415.4 (-751.1 - 618,889.8)	280,202.9 (-33,891.9 - 16,152,481)	51,537 (-1,370 - 442,552,550)
1000	0.75	0 (0 - 72.3)	0 (0 - 5,844)	0 (0 - 43)	0.561 (0 - 14.058)	0.002 (0 - 0.188)	6,420 (-115.1 - 487,665.4)	520,661.4 (-26,907.8 - 38,331,662)	70,576.5 (-532 - 316,234,239)

1	0.9	73.9 (0 - 97)	139 (0 - 7,530)	1 (0 - 76)	0.516 (0 - 19.653)	0.003 (0 - 0.389)	6,426.9 (- 1,244.7 - 547,435.8)	204,077.4 (- 42,842.7 - 9,873,671)	39,512.9 (- 1,461.1 - 356,454,086)
50	0.9	36.3 (0 - 93.3)	73 (0 - 7,497)	0 (0 - 73)	0.404 (0 - 17.636)	0.001 (0 - 0.261)	7,335.2 (-875.7 - 572,086.8)	233,928.9 (- 44,189.1 - 11,848,752)	64,389.6 (- 1,396.7 - 415,738,604)
100	0.9	17.5 (0 - 91.5)	38 (0 - 7,352)	0 (0 - 71)	0.437 (0 - 20.162)	0.001 (0 - 0.266)	6,802.5 (-700.6 - 671,894)	245,096.9 (- 43,119.9 - 13,696,372)	61,178 (- 1,451.7 - 462,146,272)
500	0.9	0 (0 - 83)	0 (0 - 6,792)	0 (0 - 57)	0.499 (0 - 13.989)	0.001 (0 - 0.14)	6,324.9 (-516.3 - 704,708.6)	329,407.8 (- 31,248.8 - 18,998,987)	60,040.6 (- 1,165 - 501,343,439)
1000	0.9	0 (0 - 75.7)	0 (0 - 6,155)	0 (0 - 45)	0.486 (0 - 12.189)	0.002 (0 - 0.167)	7,502.9 (-60.7 - 543,136.8)	600,903.9 (- 23,509.5 - 41,590,463)	81,004.7 (- 182.9 - 370,297,938)

Table S3.

Cases and deaths averted, OCV efficiency and costs in suspected cholera outbreaks observed in sub-Saharan Africa, 2010-2022. Vaccine coverage was set at 75% and vaccination was assumed to occurred at week 9 since the start of the outbreak. Outbreak data subsets were sampled randomly or weighted by cumulative cases such that the total population size matches approximately 84 million doses of OCVs that have been deployed from 2013 -2022 through OCV stockpile.

Outbreak data subset	% case reduction	Case averted	Death averted	Case averted per 1,000 OCVs	Death averted per 1,000 OCVs	Cost per case averted	Cost per death averted	Cost per DALY averted
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Random	33.4 (19.2 - 45.3)	27,649 (13,621 - 42,365)	478 (199 - 843)	0.63 (0.346 - 0.932)	0.011 (0.005 - 0.019)	5,491.1 (3,325.8 - 10,605.4)	316,528 (155,153.4 - 732,575.7)	13,071.4 (6,512.8 - 30,290.8)
Case weighted	32.7 (20 - 43.7)	44,054 (24,776 - 64,498)	640 (332 - 1,061)	0.915 (0.551 - 1.315)	0.013 (0.007 - 0.022)	3,624.6 (2,225.9 - 6,446.3)	249,112.1 (124,436.5 - 501,931.1)	10,324.8 (5,225.1 - 20,786.6)

Table S4.

Cases and deaths averted, OCV efficiency and costs in confirmed cholera outbreaks observed in sub-Saharan Africa, 2010-2023 for time-triggered vaccination scenarios. Metrics were assessed for individual outbreaks.

Campaign week	Vaccine coverage	% case reduction	Case averted	Death averted	Case averted per 1,000 OCVs	Death averted per 1,000 OCVs	Cost per case averted	Cost per death averted	Cost per DALY averted
1	0.6	66.7 (9.7 - 93.4)	214 (4 - 10,456)	2 (0 - 98)	0.675 (0.007 - 20.831)	0.003 (0 - 0.417)	6,042.3 (-784.3 - 436,616.6)	434,149 (-40,473.8 - 9,656,290)	69,308.5 (-1,787.8 - 238,471,207)
5	0.6	27 (0 - 81.9)	92 (0 - 9,598)	0 (0 - 87)	0.303 (0 - 17.8)	0 (0 - 0.252)	11,502.5 (-66.6 - 1,030,715)	684,043.8 (-38,842.4 - 11,479,787)	202,079.7 (-333.5 - 794,593,397)

9	0.6	4.7 (0 - 77.7)	15 (0 - 9,098)	0 (0 - 84)	0.136 (0 - 13.951)	0 (0 - 0.15)	21,813.1 (- 64.8 - 4,128,355)	682,393.4 (- 37,212 - 29,344,575)	694,752.5 (- 280.7 - 3,456,119,939)
13	0.6	0 (0 - 72.6)	0 (0 - 8,490)	0 (0 - 82)	0.099 (0 - 7.271)	0 (0 - 0.142)	25,337.6 (- 71.2 - 2,400,524)	393,814.4 (- 39,628.5 - 42,467,418)	768,148.3 (- 268.1 - 966,289,035)
17	0.6	0 (0 - 66.1)	0 (0 - 7,766)	0 (0 - 68)	0.139 (0 - 5.461)	0 (0 - 0.256)	14,758.4 (- 66.8 - 1,727,974)	480,779.9 (- 7,708.2 - 43,368,572)	1,010,807 (- 168.5 - 1,384,879,116)
1	0.75	74.5 (10.8 - 96.2)	235 (5 - 11,481)	2 (0 - 104)	0.592 (0.006 - 17.828)	0.002 (0 - 0.36)	6,845.2 (-782 - 496,739.9)	500,221.9 (- 40,437.3 - 10,779,882)	79,978.7 (- 1,786.2 - 255,823,069)
5	0.75	29.8 (0 - 87.9)	102 (0 - 10,483)	0 (0 - 93)	0.268 (0 - 15.169)	0 (0 - 0.225)	13,127.2 (-28 - 1,142,737)	784,822.6 (- 38,687.8 - 12,709,439)	234,015.6 (- 192 - 899,656,759)
9	0.75	5.2 (0 - 83.5)	17 (0 - 9,955)	0 (0 - 93)	0.119 (0 - 12.128)	0 (0 - 0.13)	25,128.3 (- 19.7 - 4,887,294)	787,185 (- 37,058.9 - 35,599,612)	790,227.5 (- 91.6 - 4,118,538,793)
13	0.75	0 (0 - 78.6)	0 (0 - 9,277)	0 (0 - 91)	0.087 (0 - 6.223)	0 (0 - 0.12)	28,530.9 (- 61.5 - 2,901,425)	443,037.5 (- 36,520.9 - 52,400,429)	863,030.7 (- 145.8 - 1,065,098,562)
17	0.75	0 (0 - 72.1)	0 (0 - 8,491)	0 (0 - 80)	0.123 (0 - 4.699)	0 (0 - 0.231)	16,958.1 (- 60.9 - 1,965,999)	560,282.8 (- 5,199 - 46,584,451)	1,113,895 (- 98.3 - 1,566,916,535)
1	0.9	78 (11.4 - 97.7)	244 (5 - 11,834)	2 (0 - 107)	0.509 (0.005 - 15.356)	0.002 (0 - 0.315)	7,968.8 (- 581.3 - 580,602.2)	586,461.2 (- 40,394.6 - 12,208,412)	94,735.8 (- 1,781.5 - 285,479,209)

5	0.9	31.5 (0 - 90.8)	106 (0 - 10,863)	0 (0 - 96)	0.233 (0 - 12.87)	0 (0 - 0.195)	14,987.5 (-8.1 - 1,288,647)	922,905.3 (-38,673.9 - 14,533,437)	272,529.8 (-100.8 - 1,048,434,751)
9	0.9	5.5 (0 - 87.1)	17 (0 - 10,315)	0 (0 - 96)	0.103 (0 - 10.525)	0 (0 - 0.114)	29,196.7 (24.1 - 5,738,596)	900,991.3 (-36,846.9 - 41,315,914)	896,946.5 (178.3 - 4,833,118,168)
13	0.9	0 (0 - 81.5)	0 (0 - 9,626)	0 (0 - 95)	0.075 (0 - 5.407)	0 (0 - 0.101)	33,195.4 (-53.5 - 3,415,734)	522,620.5 (-31,646.8 - 62,242,026)	994,574.9 (-56.9 - 1,209,829,683)
17	0.9	0 (0 - 74.2)	0 (0 - 8,730)	0 (0 - 88)	0.106 (0 - 4.062)	0 (0 - 0.204)	19,666.5 (-36 - 2,208,903)	618,235 (-2,291.8 - 50,855,423)	1,268,500 (-22.9 - 1,791,145,646)

Table S5.

Cases and deaths averted, OCV efficiency and costs in confirmed cholera outbreaks observed in sub-Saharan Africa, 2010-2023 for case-triggered vaccination scenarios. Metrics were assessed for individual outbreaks.

Case-trigger	Vaccine coverage	% case reduction	Case averted	Death averted	Case averted per 1,000 OCVs	Death averted per 1,000 OCVs	Cost per case averted	Cost per death averted	Cost per DALY averted
1	0.6	66.7 (9.7 - 93.4)	214 (4 - 10,456)	2 (0 - 98)	0.675 (0.007 - 20.831)	0.003 (0 - 0.417)	6,042.3 (-784.3 - 436,616.6)	434,149 (-40,473.8 - 9,656,290)	69,308.5 (-1,787.8 - 238,471,207)

50	0.6	43.8 (0 - 88.5)	151 (0 - 10,456)	1 (0 - 98)	0.54 (0 - 21.431)	0.002 (0 - 0.316)	7,937.6 (- 72.9 - 747,560.8)	523,049.6 (- 39,293.3 - 9,785,969)	91,285.8 (-476.2 - 492,708,253)
100	0.6	33.8 (0 - 86.5)	116 (0 - 10,451)	0 (0 - 98)	0.571 (0 - 22.998)	0.002 (0 - 0.313)	7,096.8 (- 72.9 - 684,854.3)	578,712.3 (- 38,869 - 10,902,670)	89,718.9 (-463.8 - 379,576,896)
500	0.6	0 (0 - 78.2)	0 (0 - 9,941)	0 (0 - 86)	0.652 (0 - 12.145)	0.005 (0 - 0.288)	6,572.3 (- 73.5 - 479,430.7)	586,547.1 (- 9,623.9 - 13,846,676)	42,115.6 (-339 - 404,333,765)
1000	0.6	0 (0 - 74.4)	0 (0 - 9,675)	0 (0 - 81)	0.576 (0 - 8.864)	0.004 (0 - 0.302)	7,894.2 (- 77.3 - 630,104.1)	730,192.8 (- 40,877.4 - 57,884,589)	47,491.8 (-372.9 - 531,406,088)
1	0.75	74.5 (10.8 - 96.2)	235 (5 - 11,481)	2 (0 - 104)	0.592 (0.006 - 17.828)	0.002 (0 - 0.36)	6,845.2 (- 782 - 496,739.9)	500,221.9 (- 40,437.3 - 10,779,882)	79,978.7 (- 1,786.2 - 255,823,069)
50	0.75	48.6 (0 - 93.2)	167 (0 - 11,481)	1 (0 - 104)	0.477 (0 - 19.002)	0.002 (0 - 0.275)	9,119.6 (- 65.8 - 839,291.8)	582,843.1 (- 39,252 - 10,831,199)	102,663.6 (- 286.9 - 549,970,601)
100	0.75	37.6 (0 - 92)	127 (0 - 11,481)	0 (0 - 104)	0.508 (0 - 19.958)	0.002 (0 - 0.273)	7,977.7 (- 68.7 - 791,570.3)	647,666.4 (- 38,793.8 - 12,204,546)	100,379.3 (- 278.4 - 422,885,301)
500	0.75	0 (0 - 85)	0 (0 - 10,926)	0 (0 - 93)	0.572 (0 - 10.468)	0.004 (0 - 0.257)	7,473.1 (-69 - 536,257.4)	670,981.4 (- 6,942 - 16,118,918)	46,912.1 (- 265.2 - 452,259,270)

1000	0.75	0 (0 - 81.9)	0 (0 - 10,592)	0 (0 - 89)	0.508 (0 - 7.538)	0.004 (0 - 0.267)	8,966.2 (- 69.3 - 724,424.4)	819,297 (- 38,444 - 69,923,984)	53,772 (-273 - 610,952,254)
1	0.9	78 (11.4 - 97.7)	244 (5 - 11,834)	2 (0 - 107)	0.509 (0.005 - 15.356)	0.002 (0 - 0.315)	7,968.8 (- 581.3 - 580,602.2)	586,461.2 (- 40,394.6 - 12,208,412)	94,735.8 (- 1,781.5 - 285,479,209)
50	0.9	51.2 (0 - 95.6)	174 (0 - 11,834)	1 (0 - 107)	0.418 (0 - 16.658)	0.001 (0 - 0.234)	10,471.8 (- 36.4 - 978,919.2)	661,352 (- 39,178.8 - 12,302,506)	117,388 (- 177 - 626,174,195)
100	0.9	39.6 (0 - 94.6)	133 (0 - 11,834)	0 (0 - 107)	0.446 (0 - 17.109)	0.001 (0 - 0.233)	9,125.2 (- 64.6 - 907,329.1)	755,710.4 (- 38,709 - 14,325,842)	115,096.5 (- 172.2 - 466,980,356)
500	0.9	0 (0 - 88)	0 (0 - 11,349)	0 (0 - 96)	0.493 (0 - 9.087)	0.004 (0 - 0.225)	8,629.1 (- 62.8 - 598,228.4)	768,508.7 (- 4,265.3 - 19,002,786)	53,590.9 (- 161.7 - 504,523,297)
1000	0.9	0 (0 - 84.9)	0 (0 - 11,024)	0 (0 - 92)	0.44 (0 - 6.812)	0.003 (0 - 0.229)	10,385.3 (- 60.4 - 838,775.9)	900,104 (- 34,433.5 - 83,240,147)	60,457.4 (- 164.4 - 707,391,955)

Table S6.

Cases and deaths averted, OCV efficiency and costs in suspected cholera outbreaks that are longer than or equal to 7 weeks observed in sub-Saharan Africa, 2010-2023 for time-triggered vaccination scenarios. Metrics were assessed for individual outbreaks.

Campaign week	Vaccine coverage	% case reduction	Case averted	Death averted	Case averted per 1,000 OCVs	Death averted per 1,000 OCVs	Cost per case averted	Cost per death averted	Cost per DALY averted
1	0.6	67.6 (23.3 - 93)	181 (13 - 7,983)	2 (0 - 82)	1.025 (0.03 - 33.797)	0.008 (0 - 0.589)	3,200.1 (-1,978.8 - 134,768.2)	132,844.8 (-51,689.1 - 6,502,663)	17,202.8 (-1,990.3 - 68,093,998)
5	0.6	29.6 (0 - 79.4)	69 (0 - 6,986)	0 (0 - 63)	0.395 (0 - 20.302)	0 (0 - 0.287)	8,744.1 (-783.7 - 810,758.5)	192,336.9 (-44,264.5 - 9,989,732)	222,838 (-1,315.9 - 564,957,947)
9	0.6	7.9 (0 - 70.7)	17 (0 - 6,092)	0 (0 - 45)	0.165 (0 - 13.95)	0 (0 - 0.198)	12,205.1 (-520.3 - 968,639.3)	223,015.2 (-40,153.7 - 13,781,393)	980,373.4 (-1,110.1 - 740,093,735)
13	0.6	0 (0 - 62.2)	0 (0 - 4,706)	0 (0 - 34)	0.122 (0 - 11.035)	0 (0 - 0.152)	15,941.4 (-379.6 - 1,342,322)	255,466.3 (-24,148.7 - 15,274,743)	2,080,830 (-577.5 - 964,149,939)
17	0.6	0 (0 - 53.2)	0 (0 - 3,884)	0 (0 - 18)	0.13 (0 - 7.921)	0 (0 - 0.144)	15,067.5 (-285 - 1,389,936)	263,129 (-23,565.5 - 19,257,356)	1,606,477 (-294.2 - 1,081,417,804)
1	0.75	75.5 (25.8 - 95.7)	200 (14 - 8,737)	2 (0 - 91)	0.905 (0.027 - 29.809)	0.007 (0 - 0.516)	3,704.6 (-1,702.3 - 150,658.8)	155,793.6 (-48,850 - 7,360,415)	19,928.1 (-1,841.3 - 77,142,415)
5	0.75	32.7 (0 - 85.1)	77 (0 - 7,648)	0 (0 - 70)	0.349 (0 - 17.918)	0 (0 - 0.252)	10,008.1 (-701.8 - 921,328.5)	222,834.1 (-40,541.1 - 11,184,469)	260,248.6 (-1,065 - 619,264,601)

9	0.75	8.8 (0 - 76.3)	19 (0 - 6,699)	0 (0 - 49)	0.146 (0 - 12.086)	0 (0 - 0.175)	14,042.7 (- 447.1 - 1,089,603)	259,129.7 (- 36,133.8 - 15,611,394)	1,130,466 (- 870.4 - 818,044,413)
13	0.75	0 (0 - 67.4)	0 (0 - 5,178)	0 (0 - 38)	0.108 (0 - 9.715)	0 (0 - 0.131)	18,002.9 (- 285.4 - 1,506,539)	295,234.2 (- 18,134.2 - 17,207,080)	2,384,529 (- 398.2 - 1,072,901,744)
17	0.75	0 (0 - 58.4)	0 (0 - 4,275)	0 (0 - 20)	0.114 (0 - 6.958)	0 (0 - 0.124)	17,131.6 (- 126.5 - 1,592,361)	304,090.3 (- 15,524.9 - 23,296,291)	1,806,850 (- 171 - 1,227,312,224)
1	0.9	78.9 (27.8 - 97.3)	209 (15 - 9,057)	2 (0 - 94)	0.789 (0.024 - 25.927)	0.006 (0 - 0.448)	4,363.4 (- 1,454.9 - 170,155.5)	183,493.5 (- 45,397 - 8,347,965)	23,448 (- 1,650 - 88,177,606)
5	0.9	34.2 (0 - 87.8)	80 (0 - 8,019)	0 (0 - 72)	0.303 (0 - 15.539)	0 (0 - 0.218)	11,562.5 (- 518.1 - 1,068,429)	259,714.3 (- 37,657 - 12,987,377)	305,317.1 (- 1,036.6 - 694,385,762)
9	0.9	9.2 (0 - 78.7)	20 (0 - 6,962)	0 (0 - 51)	0.128 (0 - 10.481)	0 (0 - 0.151)	16,330.6 (- 354.6 - 1,259,715)	304,272.2 (- 30,946.7 - 18,139,001)	1,308,735 (- 727.6 - 914,158,096)
13	0.9	0 (0 - 69.7)	0 (0 - 5,387)	0 (0 - 39)	0.094 (0 - 8.388)	0 (0 - 0.113)	20,645.5 (-99.6 - 1,719,769)	344,613.8 (- 15,980.8 - 19,841,520)	2,758,140 (- 173.9 - 1,216,690,912)
17	0.9	0 (0 - 60.5)	0 (0 - 4,433)	0 (0 - 22)	0.1 (0 - 6.087)	0 (0 - 0.11)	19,789.9 (-40.2 - 1,818,713)	358,109.2 (- 14,885.2 - 26,752,721)	2,061,254 (- 38.3 - 1,417,275,879)

Table S7.

Cases and deaths averted, OCV efficiency and costs in suspected cholera outbreaks that are longer than or equal to 7 weeks observed in sub-Saharan Africa, 2010-2023 for case-triggered vaccination scenarios. Metrics were assessed for individual outbreaks.

Case-trigger	Vaccine coverage	% case reduction	Case averted	Death averted	Case averted per 1,000 OCVs	Death averted per 1,000 OCVs	Cost per case averted	Cost per death averted	Cost per DALY averted
1	0.6	67.5 (23 - 93)	181 (12 - 7,983)	2 (0 - 82)	1.024 (0.03 - 33.797)	0.008 (0 - 0.589)	3,202.9 (-1,978.8 - 135,519.5)	133,114.1 (-51,711.3 - 6,505,950)	17,308.9 (-1,990.3 - 68,688,749)
50	0.6	41.4 (0 - 86.4)	120 (0 - 7,906)	1 (0 - 80)	0.679 (0 - 26.727)	0.003 (0 - 0.387)	4,861.4 (-1,293 - 331,304.8)	164,903.5 (-50,269.9 - 8,127,173)	37,904 (-1,859.8 - 234,172,124)
100	0.6	27.1 (0 - 84)	80 (0 - 7,840)	0 (0 - 79)	0.685 (0 - 28.878)	0.003 (0 - 0.385)	4,717.3 (-1,202.7 - 454,942.7)	174,661.2 (-49,097.3 - 9,696,733)	40,778.2 (-1,856.9 - 323,565,430)
500	0.6	0 (0 - 75.6)	0 (0 - 7,400)	0 (0 - 62)	0.71 (0 - 18.526)	0.002 (0 - 0.195)	4,529.9 (-821.4 - 479,060.8)	238,371.5 (-36,781.2 - 13,134,258)	44,587.9 (-1,459.6 - 368,221,270)
1000	0.6	0 (0 - 69.3)	0 (0 - 6,905)	0 (0 - 49)	0.682 (0 - 16.405)	0.002 (0 - 0.212)	5,406.4 (-189.4 - 330,119.5)	453,707.6 (-28,739.5 - 26,797,729)	61,099.1 (-797.2 - 260,543,049)
1	0.75	75.4 (25.6 - 95.7)	200 (14 - 8,737)	2 (0 - 91)	0.904 (0.027 - 29.809)	0.007 (0 - 0.516)	3,709 (-1,702.3 - 151,393.7)	156,219.6 (-48,870.6 - 7,378,561)	20,048.7 (-1,841.3 - 77,740,420)
50	0.75	46.1 (0 - 91.7)	132 (0 - 8,656)	1 (0 - 89)	0.598 (0 - 23.519)	0.003 (0 - 0.34)	5,578.8 (-1,128.3 - 376,140.7)	192,912.9 (-47,147.2 - 9,369,476)	43,507.8 (-1,717.3 - 268,975,130)

100	0.75	30 (0 - 89.9)	89 (0 - 8,605)	0 (0 - 87)	0.603 (0 - 25.353)	0.002 (0 - 0.334)	5,422.5 (-979 - 515,278.4)	202,367.8 (-46,341.9 - 10,904,716)	46,822.5 (-1,728.7 - 368,186,764)
500	0.75	0 (0 - 82.3)	0 (0 - 8,146)	0 (0 - 68)	0.62 (0 - 16.444)	0.002 (0 - 0.172)	5,245.4 (-755.6 - 543,610.4)	278,783.6 (-34,126 - 14,754,305)	50,828.6 (-1,375.5 - 415,319,276)
1000	0.75	0 (0 - 76.3)	0 (0 - 7,594)	0 (0 - 54)	0.595 (0 - 14.156)	0.002 (0 - 0.193)	6,248.7 (-115.3 - 374,298.7)	519,677 (-27,034.8 - 31,552,548)	69,649.5 (-540.5 - 301,025,522)
1	0.9	78.8 (27.5 - 97.3)	209 (15 - 9,057)	2 (0 - 94)	0.788 (0.024 - 25.927)	0.006 (0 - 0.448)	4,366.2 (-1,454.9 - 171,665)	183,906 (-45,486 - 8,353,269)	23,543.6 (-1,650 - 88,801,470)
50	0.9	48.4 (0 - 94.2)	139 (0 - 8,998)	1 (0 - 93)	0.522 (0 - 20.456)	0.002 (0 - 0.293)	6,498.4 (-904.3 - 425,573.9)	229,084.9 (-44,170.2 - 10,733,393)	50,208.5 (-1,432.9 - 301,408,950)
100	0.9	31.6 (0 - 92.4)	94 (0 - 8,998)	0 (0 - 92)	0.525 (0 - 21.865)	0.002 (0 - 0.29)	6,315.4 (-715.6 - 591,520.1)	239,087.8 (-43,779.8 - 12,513,202)	54,147.1 (-1,513.2 - 427,296,718)
500	0.9	0 (0 - 85.4)	0 (0 - 8,504)	0 (0 - 71)	0.536 (0 - 14.379)	0.002 (0 - 0.149)	6,173.3 (-542.1 - 617,944.4)	327,928.3 (-31,356 - 16,982,331)	59,644.8 (-1,174.4 - 476,064,382)
1000	0.9	0 (0 - 79.6)	0 (0 - 7,898)	0 (0 - 56)	0.522 (0 - 12.219)	0.002 (0 - 0.169)	7,262.9 (-60.8 - 439,012.8)	600,441.5 (-23,642.6 - 37,363,824)	80,317.9 (-182.9 - 343,214,391)

Figure S5.

Percentage reduction in cases (A) and deaths (B), as well as cases (C) and deaths (D) averted per 1,000 OCV doses across campaign timings for four distinct subsets of outbreaks. Metrics were computed collectively across all outbreaks in a subset. Purple color represents the results based on the uniform random sample, and blue, red, and green colors represent results based on the samples weighted by attack rate, cumulative cases, and duration of an outbreak, respectively. Shaded boxes represent the central 50% of delays before campaign vaccination, as reported in the GTFCC dashboard.

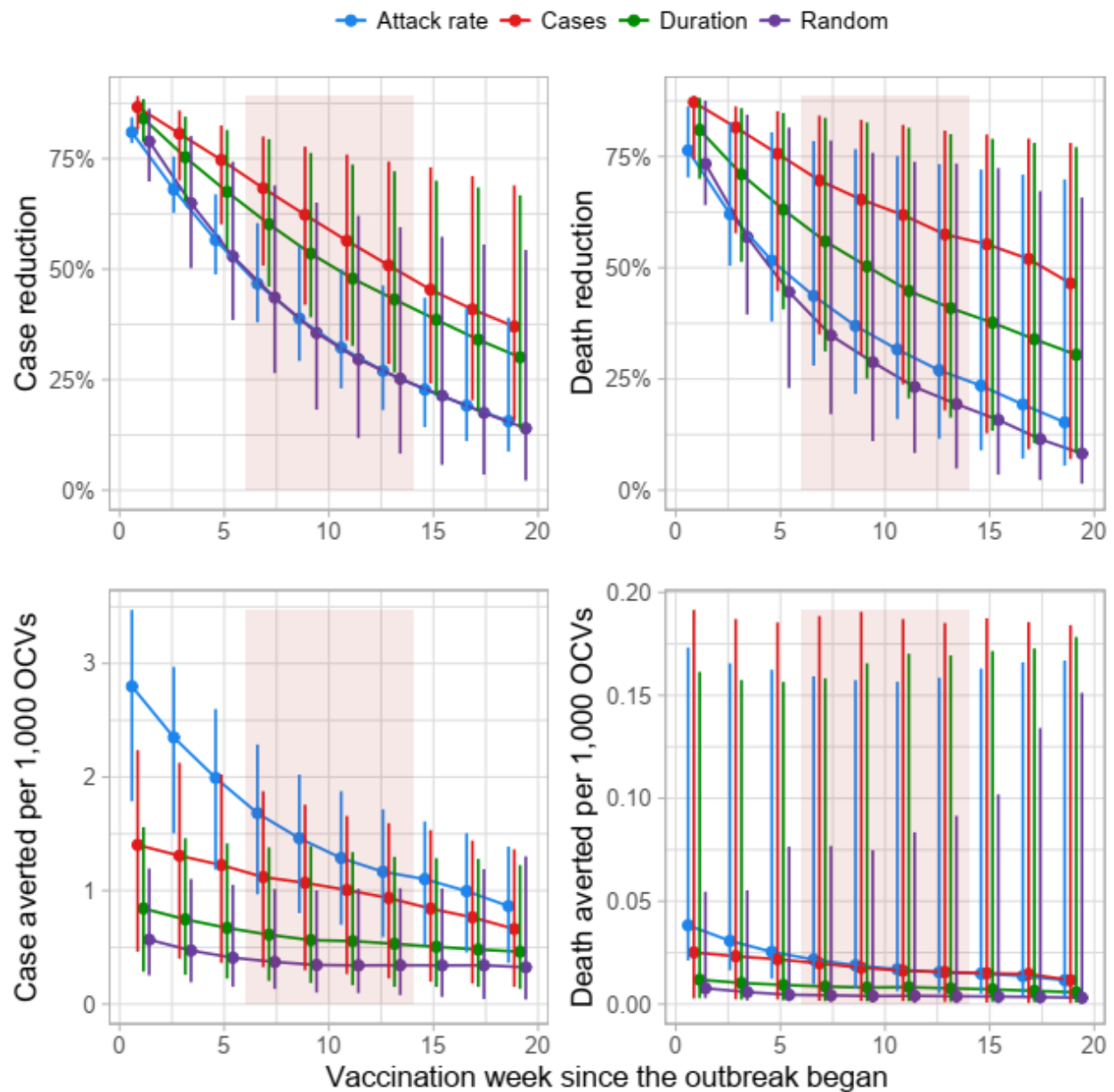


Figure S6.

Median case reduction (%) across vaccination campaign timing for outbreak subsets categorized by outbreak duration (**A** and **B**) and attack rate (**C** and **D**). Left panel (**A** and **C**) shows results under time-triggered vaccination scenario whereas right panel (**B** and **D**) shows the results under case-triggered vaccination scenarios. Y-axis label, $\geq P_{X\%}$, indicates a subset of outbreak data for which the total cases are larger than or equal to Xth percentile.

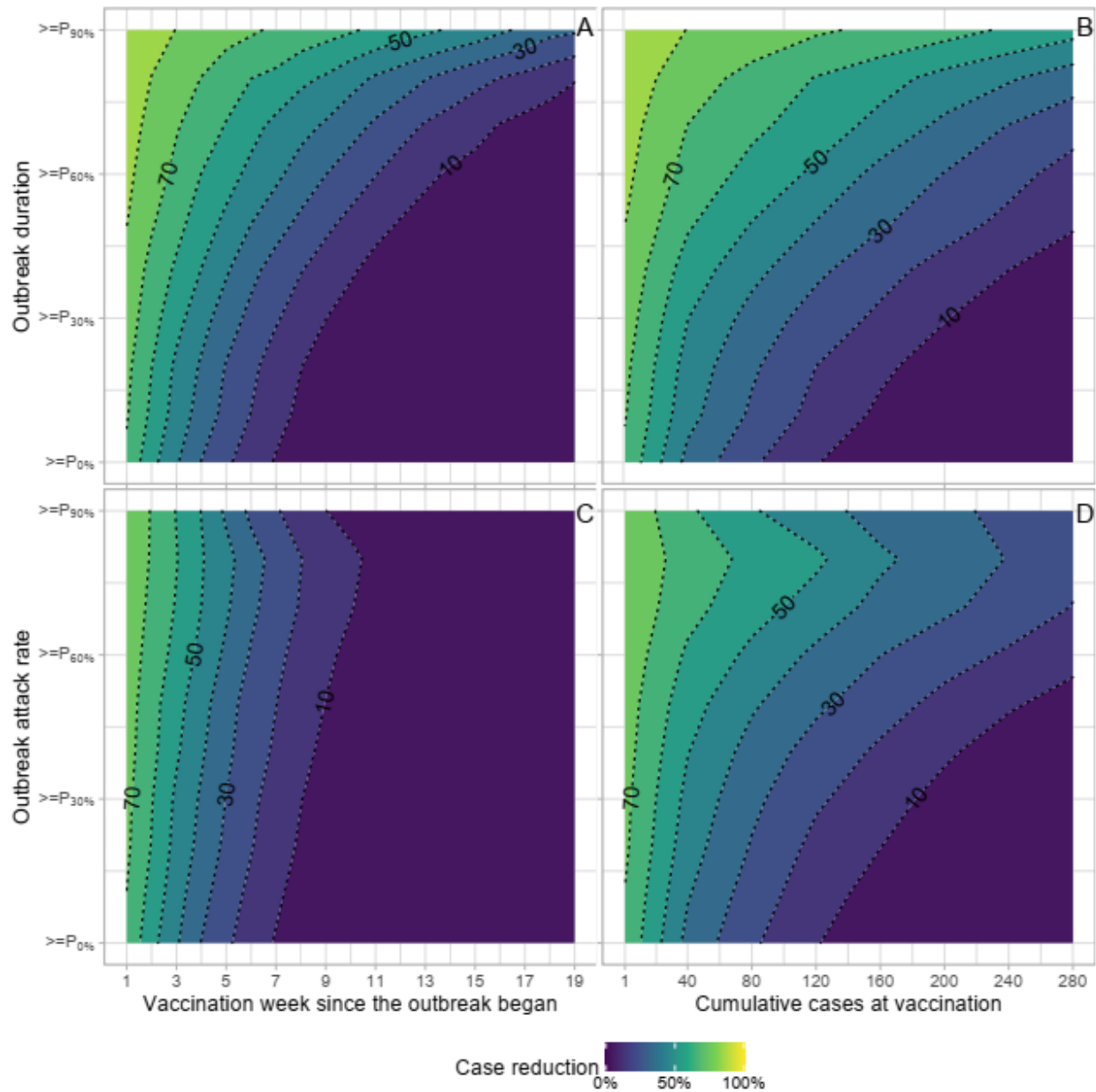


Figure S7.

Outbreak duration and total suspected cases for all outbreaks studied in the present study (blue) and those reported in the literature (red). Model and Review refer to outbreaks in our dataset (labeled “Model”) and those identified in the literature review (“Review”), respectively.

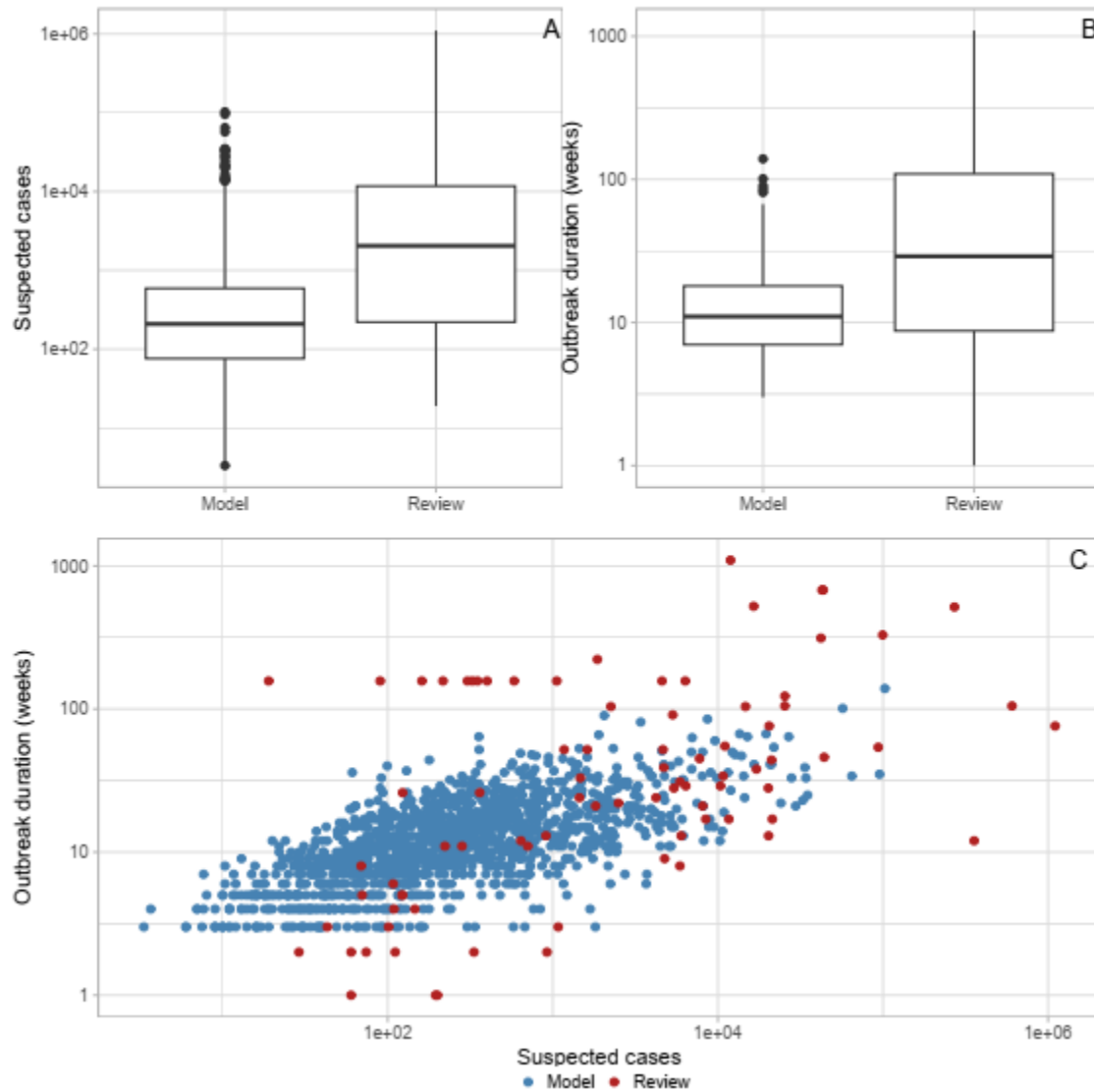


Figure S8.

Direct, indirect, total, and overall vaccine effectiveness in the dynamic and static models. Vaccine effectiveness was measured under the pre-emptive vaccination scenario where a proportion of the individuals, determined by the vaccine coverage rate, were in the compartments of individuals who acquired vaccine-derived protection, V^P (Figure S2), in the beginning of the simulation.

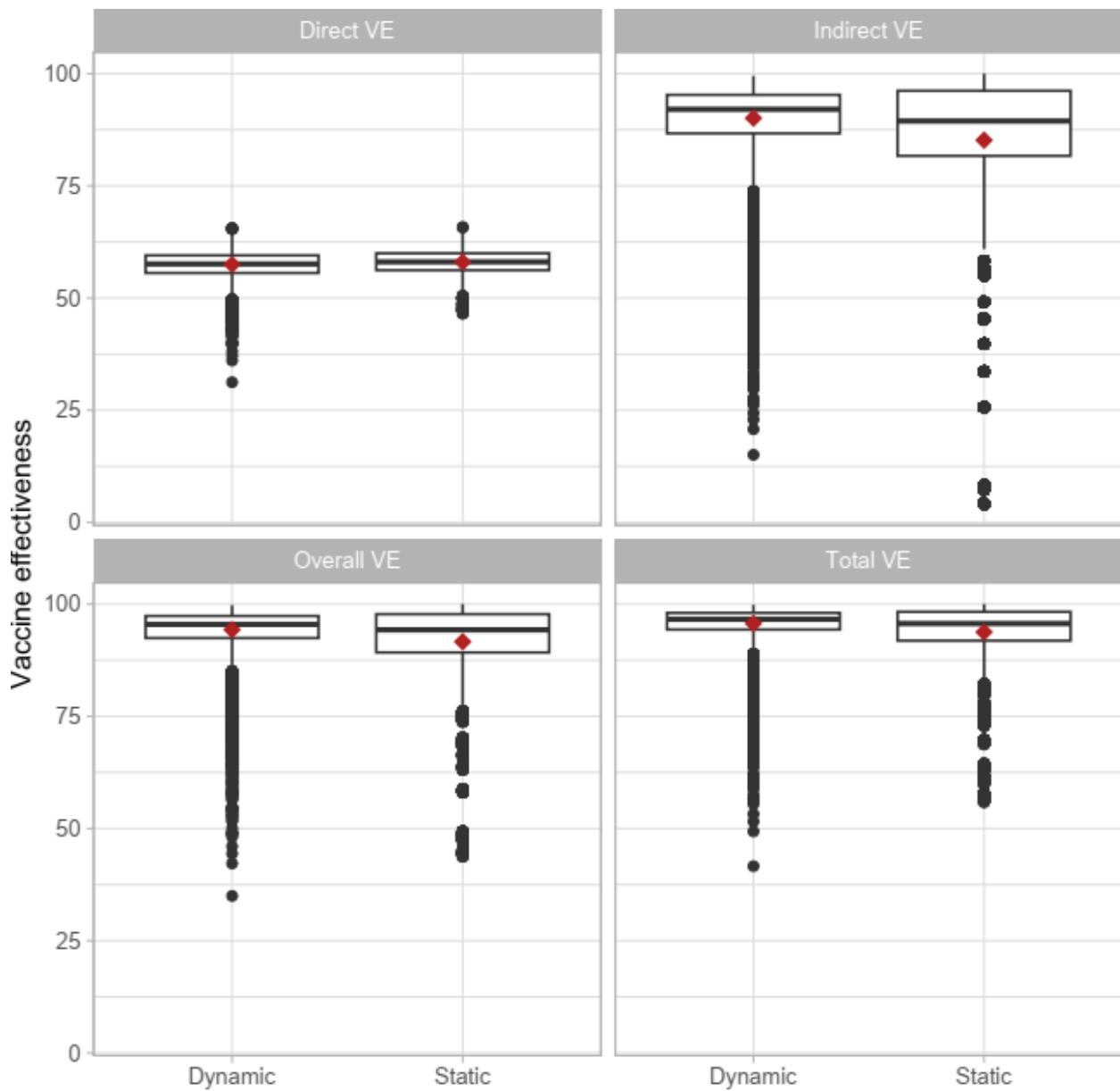
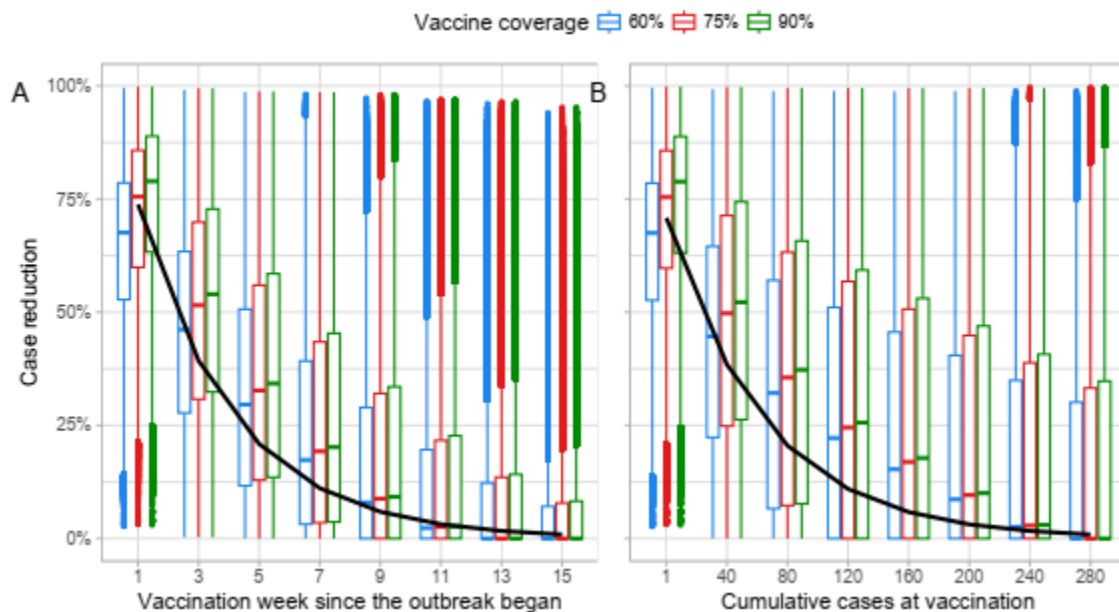


Figure S9.

Percent case reduction by the ORI during cholera outbreaks that last 7 weeks or longer. **A** and **B** show the percentage case reduction at three vaccine coverage rates (i.e., 60%, 75%, and 90%) across time and case campaign triggers. The exponential decays (black lines) indicate the percentage case reduction of 24.7% by each week's delay (halving approximate every 2.8 weeks) and 10.7% percent by 10 additional cases (halving approximately every 69 cases).



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